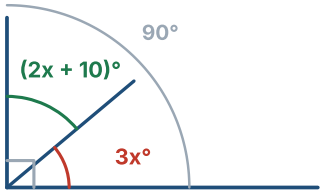
SAT Math 4.1 — Lines, Angles & Triangles

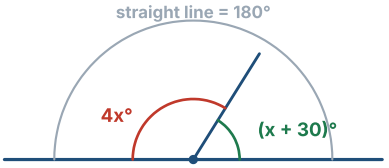
Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

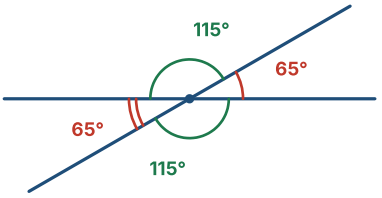
1 Angles on a line and at a point

straight line = 180° | full turn = 360° | vertical angles are equal

Complementary angles add to 90°; supplementary to 180°. Vertical (opposite) angles made by two crossing lines are always equal. Most angle-chasing questions are one of these three facts applied twice.



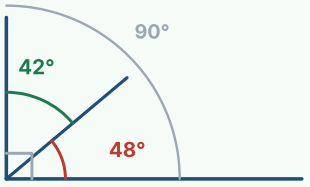




Three pictures, three facts: a right angle split in two parts that add to 90°; a straight line split in two that add to 180°; and two crossing lines, where opposite angles are equal and neighbours add to 180°.

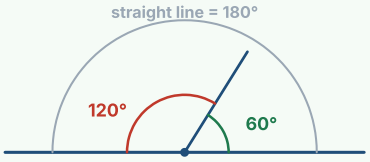
Example 1. **Two angles are complementary. One is 3x° and the other (2x + 10)°. Find x.**

Solution.
3x + 2x + 10 = 90, so 5x = 80.
x = **16**.



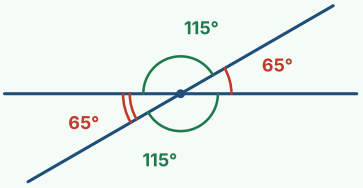
Example 2. **Two angles on a straight line are (4x)° and (x + 30)°. Find x.**

Solution.
4x + x + 30 = 180, so 5x = 150.
x = **30**.



Example 3. **Two lines cross. One angle is 65°. Find the other three.**

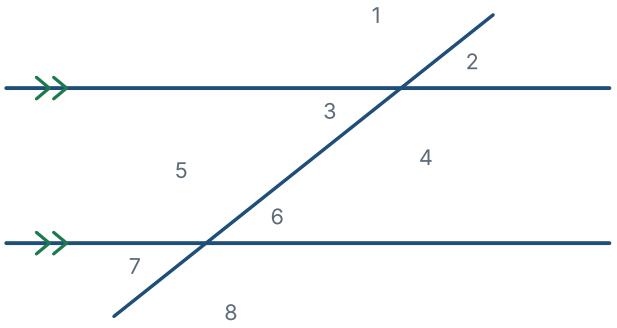
Solution.
The vertical angle is **65°**; the two adjacent ones are each 180 − 65 = **115°**.



2 Parallel lines and a transversal

corresponding and alternate angles are equal | co-interior angles add to 180°

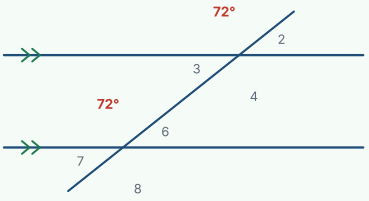
When a line crosses two parallels, every angle is either equal to a given one or supplementary to it. Find the equal pairs first; everything else follows from 180°.



The arrow marks say the two lines are parallel. Angles 1 and 5 are corresponding (equal), 4 and 5 are alternate interior (equal), and 3 and 5 are co-interior (they add to 180°).

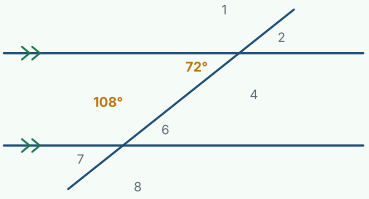
Example 1. **Two parallel lines are cut by a transversal and one angle is 72°. What is its corresponding angle?**

Solution.
Corresponding angles are equal.
72°.



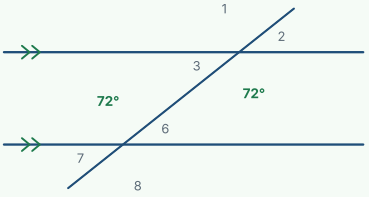
Example 2. **What is the co-interior angle to that 72°?**

Solution.
Co-interior angles add to 180°.
180 − 72 = **108°**.



Example 3. **In the figure above, angles 4 and 5 are what kind of pair?**

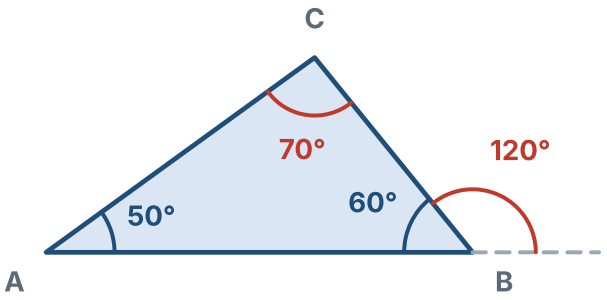
Solution.
They sit between the parallels on opposite sides of the transversal: **alternate interior**, so they are equal.



3 Triangle angle facts

angles sum to 180° | an exterior angle equals the two remote interior angles

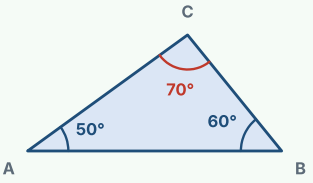
The exterior-angle shortcut saves a step: instead of finding the third angle and subtracting from 180, add the two far angles directly.



The three angles inside total 180°. The exterior angle at B is 120°, which is both 180 − 60 and 50 + 70 — the two angles far away from it.

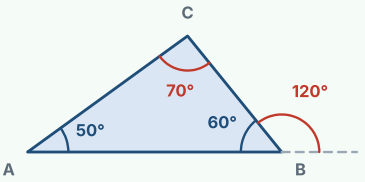
Example 1. **A triangle has angles 50° and 60°. Find the third.**

Solution.
180 − 50 − 60 = **70°**.



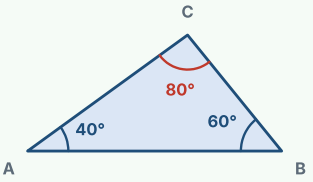
Example 2. **For that triangle, what is the exterior angle at the 60° vertex?**

Solution.
It equals the two remote interior angles: 50 + 70 = **120°**. Checking: 180 − 60 = 120.



Example 3. **A triangle's angles are in the ratio 2 : 3 : 4. Find the largest.**

Solution.
The parts total 9, and 180/9 = 20° per part.
The largest is 4 × 20 = **80°**.



4 Isosceles and equilateral triangles

equal sides face equal angles | equilateral → every angle 60°

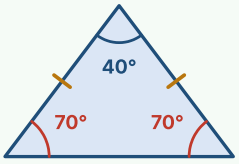
The tick marks on a diagram are doing real work: two marked sides mean two equal base angles. This turns most isosceles questions into a single equation.



Two equal sides (the tick marks) force two equal base angles, so 2x + 40 = 180. When all three sides are marked, every angle is 60°.

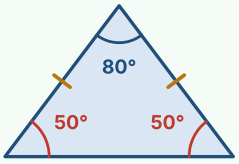
Example 1. **An isosceles triangle has apex angle 40°. Find each base angle.**

Solution.
The two base angles are equal: (180 − 40)/2.
70° each.



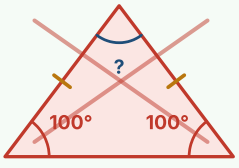
Example 2. **An isosceles triangle has a base angle of 50°. Find the apex angle.**

Solution.
Both base angles are 50°.
180 − 100 = **80°**.



Example 3. **Why can an isosceles triangle not have two angles of 100°?**

Solution.
They would total 200°, already above 180°. The equal angles in an isosceles triangle must each be **below 90°**.



5 The triangle inequality

any two sides must add to more than the third

Given two sides a and b, the third side x satisfies $|a - b| < x < a + b$. Questions ask for the possible values, or for the largest or smallest whole number that works.



Hinge the two known sides open and shut. Straightened out they reach 7 + 10 = 17; folded back they close to 10 − 7 = 3. The third side lives strictly between, which is the red stretch of the number line.

Example 1. **Two sides are 7 and 10. What values can the third side take?**

Solution.

$10 - 7 < x < 10 + 7$.

$3 < x < 17$.



Example 2. **For those sides, what is the largest possible whole-number third side?**

Solution.

x must be below 17.

16.



Example 3. **Can a triangle have sides 4, 5 and 9?**

Solution.

4 + 5 = 9, which is not *more* than 9.

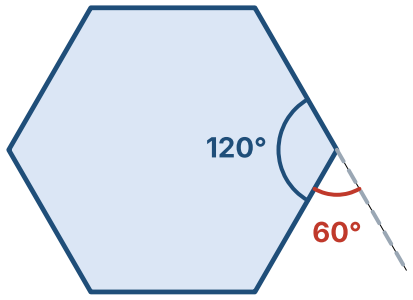
No — it collapses flat.



6 Polygons and quadrilaterals

interior angles of an n-gon sum to (n − 2) × 180° | exterior angles always sum to 360°

The exterior-angle fact is the quicker route for regular polygons: each exterior angle is 360/n, and the interior angle is its supplement. In a parallelogram opposite angles are equal and adjacent ones add to 180°.



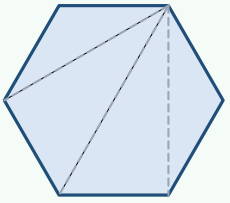
interior + exterior = 180°



Extend one side of the hexagon and the exterior angle appears: 60°, with 120° inside. Walk right round and the exterior angles total 360°. In the parallelogram, 65° sits opposite 65° and beside 115°.

Example 1. **Find the sum of the interior angles of a hexagon.**

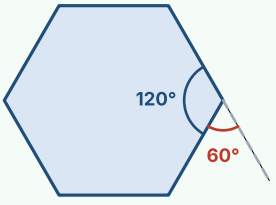
Solution.
(6 − 2) × 180 = **720°**.



4 triangles × 180° = 720°

Example 2. **Find each interior angle of a regular hexagon.**

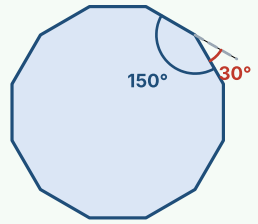
Solution.
Each exterior angle is 360/6 = 60°.
Interior = 180 − 60 = **120°**.



interior + exterior = 180°

Example 3. **A regular polygon has interior angles of 150°. How many sides?**

Solution.
Each exterior angle is 30°, and they total 360°.
360/30 = **12 sides**.



interior + exterior = 180°

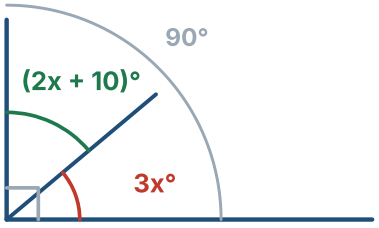
Part 1 — With Desmos (10 questions · draw the figure; the picture is faster than the algebra)

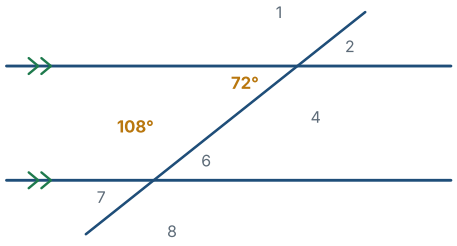
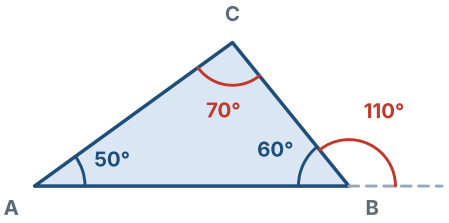
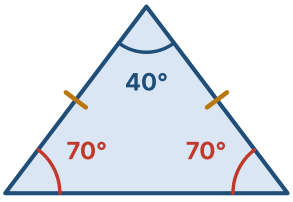
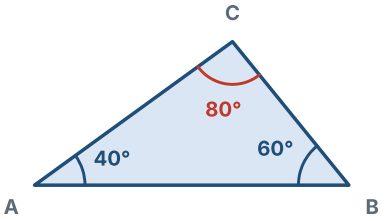
The SAT gives you Desmos on every maths question. For angle work it is **an equation solver**. Write the angle relationship as an equation and let Desmos finish it — the thinking is choosing which relationship applies.

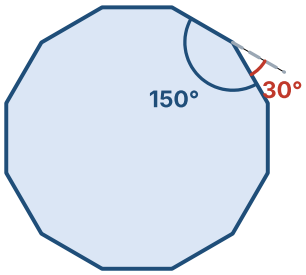
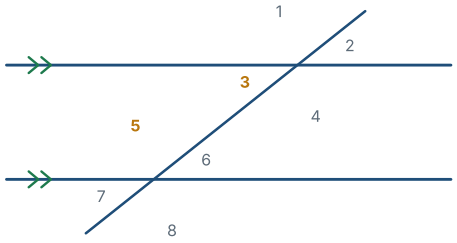
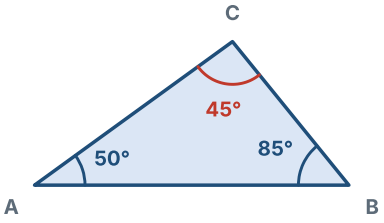
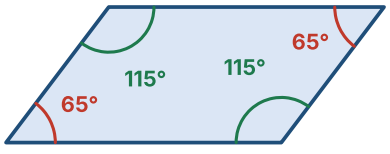
#	Question	Type into Desmos	Then	Answer
D1	Two complementary angles measure 3x° and (2x + 10)°. What is x?	`3x+2x+10=90`	Type the equation; Desmos solves it.	16
D2	Two angles on a straight line are 4x° and (x + 30)°. What is x?	`4x+x+30=180`	A straight line gives 180.	30
D3	A triangle has angles 50° and 60°. What is the third angle?	`180-50-60`	The three must total 180.	70
D4	A triangle's angles are in the ratio 2 : 3 : 4. What is the largest angle?	`180/9*4`	Nine parts in total.	80
D5	An isosceles triangle has apex angle 40°. What is each base angle?	`(180-40)/2`	The two base angles are equal.	70
D6	What is the sum of the interior angles of a hexagon?	`(6-2)*180`	The formula works for any polygon.	720
D7	What is each interior angle of a regular hexagon?	`180-360/6`	Exterior first, then subtract from 180.	120
D8	A regular polygon has interior angles of 150°. How many sides does it have?	`360/(180-150)`	Exterior angles always total 360.	12
D9	Two sides of a triangle are 7 and 10. What is the largest whole-number third side?	`7+10-1`	Strictly less than the sum.	16
D10	Two parallel lines are cut by a transversal and one angle is 72°. What is the co-interior angle?	`180-72`	Co-interior angles are supplementary.	108

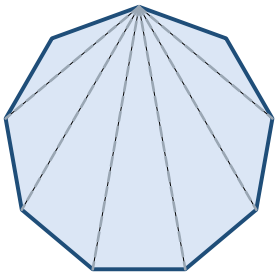
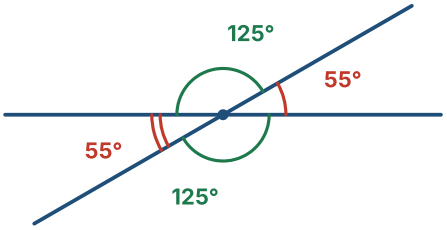
Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Mark every angle you can deduce on the figure before you write an equation. Most of these fall out in two steps once the diagram is labelled.

#	Level	Question	Answer	Solution
P1	Hard	Two angles are complementary. One angle measures 3x° and the other measures (2x + 10)°. What is the value of x? 	16	Complementary means they total 90°: 3x + 2x + 10 = 90, so 5x = 80. x = **16**. Using 180 instead of 90 gives 34, which is not an option — a useful check.

#	Level	Question	Answer	Solution
P2	Hard	In the figure, two parallel lines are cut by a transversal. If angle 1 measures 72°, what is the measure of angle 3, in degrees? 	108	Angles 1 and 3 sit on a straight line, so they are supplementary. 180 − 72 = **108°**.
P3	Hard	In a triangle, two of the angles measure 50° and 60°. What is the measure of the exterior angle at the third vertex, in degrees? 	110	The third interior angle is 180 − 110 = 70°, and its exterior angle is 180 − 70 = **110°**. Equivalently, the exterior angle equals the two remote interior angles: 50 + 60 = 110.
P4	Hard	An isosceles triangle has an apex angle of 40°. What is the measure of each base angle? 	70°	The two base angles are equal and the three total 180°. (180 − 40)/2 = **70°**. A common wrong answer is the sum of the two base angles, not each one.
P5	Hard	The angles of a triangle are in the ratio 2 : 3 : 4. What is the measure of the largest angle, in degrees? 	80	The ratio has 2 + 3 + 4 = 9 parts, and 180/9 = 20° per part. 4 × 20 = **80°**.
P6	Hard	Two sides of a triangle measure 7 and 10. State the range of possible lengths for the third side. 	3 < x < 17	The third side x must satisfy **3 < x < 17**. Both boundaries are excluded, and each would give a flat, degenerate triangle.

#	Level	Question	Answer	Solution
P7	Very hard	In a regular polygon, each interior angle measures 150°. How many sides does the polygon have?  interior + exterior = 180°	12	Each exterior angle is 180 − 150 = 30°, and the exterior angles always total 360°. 360/30 = **12 sides**.
P8	Very hard	Two parallel lines are cut by a transversal. One of the eight angles formed measures (3x + 10)° and the co-interior angle on the same side measures (2x − 5)°. What is the value of x?  	35	Co-interior angles are supplementary: 3x + 10 + 2x − 5 = 180, so 5x + 5 = 180. x = **35**. Setting them equal instead gives x = −15, which is impossible — a sign you chose the wrong angle relationship.
P9	Very hard	In triangle ABC, angle A measures (2x)°, angle B measures (3x + 10)°, and angle C measures (x + 20)°. What is the measure of the largest angle, in degrees?  	85°	2x + 3x + 10 + x + 20 = 180 gives 6x + 30 = 180, so x = 25. The angles are 50°, 85° and 45°, so the largest is **85°**. A common wrong answer is x itself, which is not an angle of the triangle.
P10	Very hard	A parallelogram has one angle measuring 65°. What is the measure of the angle adjacent to it, in degrees?  	115	Adjacent angles of a parallelogram are supplementary. 180 − 65 = **115°**. The opposite angle, by contrast, is also 65°.

#	Level	Question	Answer	Solution
P11	Very hard	The sum of the interior angles of a polygon is 1,260°. How many sides does it have?  7 triangles × 180° = 1260°	9	(n − 2) × 180 = 1260, so n − 2 = 7. n = **9**. A common wrong answer is n − 2, stopped one step early.
P12	Very hard	In the figure, two lines intersect and one of the four angles formed measures (5x − 20)°. Its vertical angle measures (3x + 10)°. What is the measure of each of these angles, in degrees? 	55°	Vertical angles are equal: 5x − 20 = 3x + 10, so 2x = 30 and x = 15. The angle is 5(15) − 20 = **55°**. A common wrong answer is x itself, not the angle — read what the question asks for.

The one habit to build. Read what the question asks for. Several of these solve to a value of x that is *not* the answer — the angle is 5x − 20, not x.

All 18 worked examples and all 22 answers checked numerically (see verify.py); every figure is drawn to the numbers in the question.

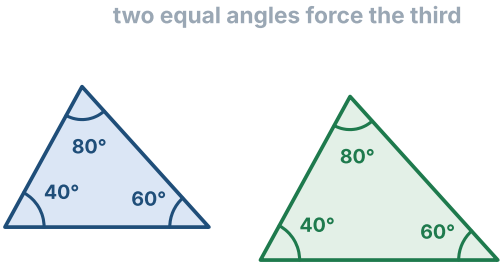
SAT Math 4.2 — Similarity & Congruence

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

1 When are two triangles similar?

AA — two pairs of equal angles is enough | similar means same shape, different size

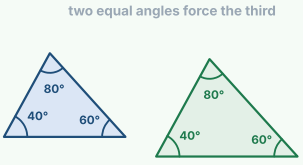
You never need all three angles: if two match, the third must. Similar triangles have equal angles and proportional sides, and the SAT almost always gives you the angles and asks for a side.



Same three angles, different sizes. Nothing here fixes the size, which is exactly why AA proves similarity but never congruence.

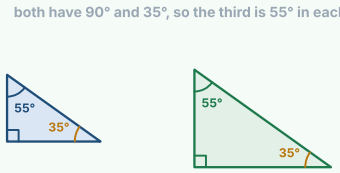
Example 1. **Two triangles have angles 40°, 60° and 40°, 80°. Are they similar?**

Solution.
The first has third angle 80°, the second 60°. Both are 40, 60, 80.
Yes, by AA.



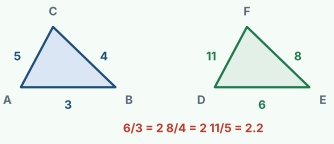
Example 2. **Two right triangles each have an angle of 35°. Are they similar?**

Solution.
Both have 90° and 35°, so the third is 55° in each.
Yes, by AA.



Example 3. **Two triangles have sides 3, 4, 5 and 6, 8, 11. Are they similar?**

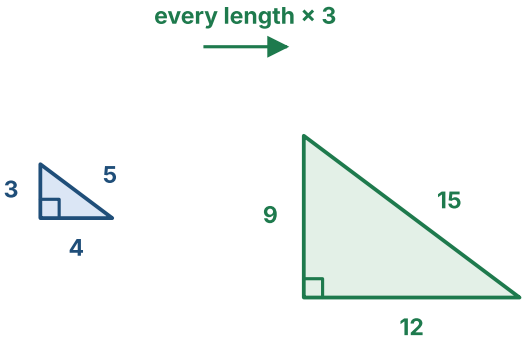
Solution.
6/3 = 2 and 8/4 = 2, but 11/5 = 2.2.
No — the ratios must all match.



2 The scale factor

scale factor k = new ÷ old | every matching length multiplies by k

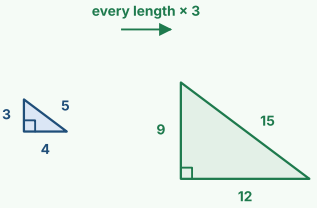
Find k from the one pair of sides you are given, then apply it to the side you want. Pair the sides correctly: the side opposite the 30° angle in one triangle matches the side opposite the 30° angle in the other.



One pair of matching sides gives k = 9/3 = 3. From there every other length follows: 4 → 12 and 5 → 15.

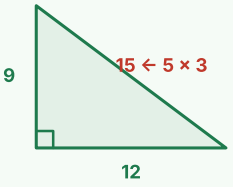
Example 1. **A triangle with sides 3, 4, 5 is enlarged so the shortest side becomes 9. Find the scale factor.**

Solution.
k = 9/3 = **3**.



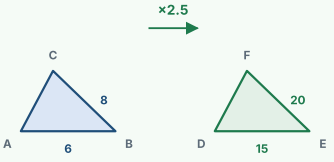
Example 2. **For that enlargement, what is the new longest side?**

Solution.
5 × 3 = **15**.



Example 3. **Triangle ABC has AB = 6; similar triangle DEF has DE = 15. If BC = 8, find EF.**

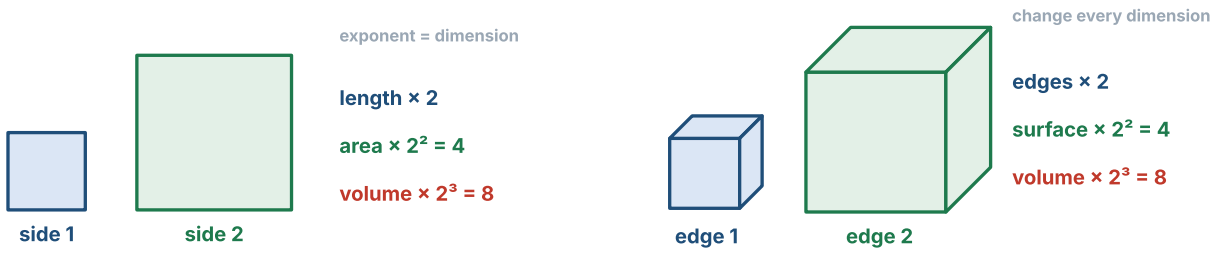
Solution.
k = 15/6 = 2.5.
EF = 8 × 2.5 = **20**.



3 Area and volume scale differently

lengths × k | areas × k^2 | volumes × k^3

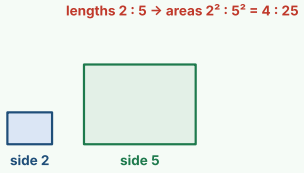
This is the trap the SAT returns to. Doubling every length multiplies the area by 4 and the volume by 8. Going the other way, if areas are in the ratio 9 : 4, the lengths are in the ratio 3 : 2 — take the square root.



Double the side and four copies of the small square fit inside the big one; eight copies of the small cube fit inside the big cube. The exponent matches the dimension: 1 for length, 2 for area, 3 for volume.

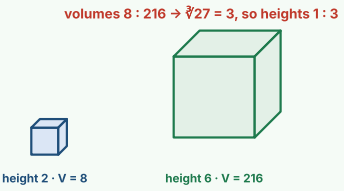
Example 1. **Two similar rectangles have sides in the ratio 2 : 5. What is the ratio of their areas?**

Solution.
Square the factor.
4 : 25.



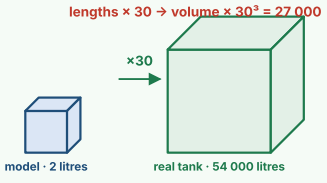
Example 2. **Two similar solids have volumes 8 cm³ and 216 cm³. Find the ratio of their heights.**

Solution.
216/8 = 27, and the cube root of 27 is 3.
1 : 3.



Example 3. **A model is built to scale 1 : 30. The model holds 2 litres. How much does the real tank hold?**

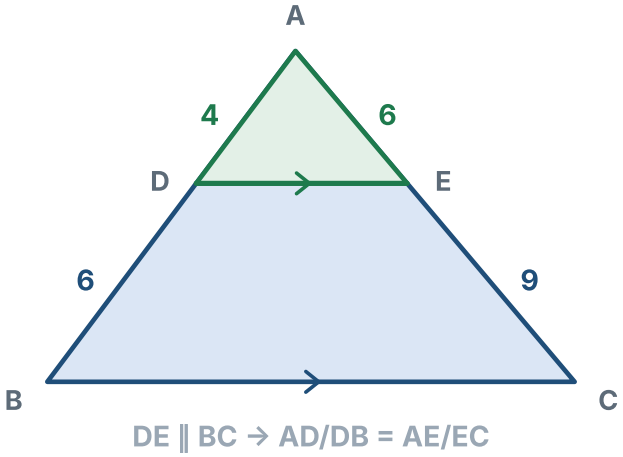
Solution.
Volume scales by 30^3 = 27,000.
2 × 27,000 = **54,000 litres**.



4 Similar triangles hidden in one figure

a line parallel to one side cuts off a similar triangle

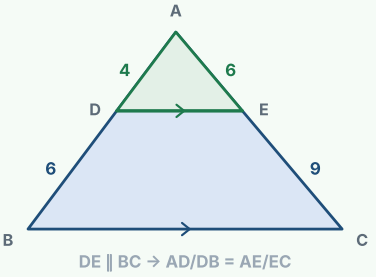
When a figure contains a line parallel to a side, two similar triangles are sitting inside it. Redraw them separately, the same way round, before writing any ratio.



DE ∥ BC makes the angles at D and B equal, and the angle at A is shared, so triangle ADE is similar to triangle ABC. The matching sides give AD/DB = AE/EC.

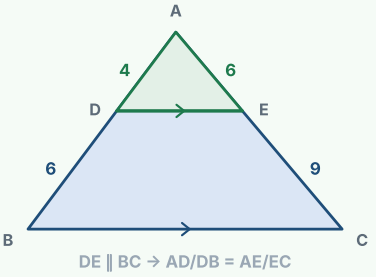
Example 1. **In triangle ABC, DE is parallel to BC with AD = 4, DB = 6 and AE = 6. Find EC.**

Solution.
AD/DB = AE/EC, so 4/6 = 6/EC.
EC = **9**.



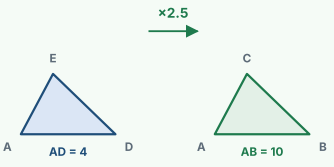
Example 2. **Why are the two triangles similar?**

Solution.
DE parallel to BC makes the corresponding angles equal, so **AA** applies.



Example 3. **In that figure, AB = 10 and AC = 15. What is the scale factor from ADE to ABC?**

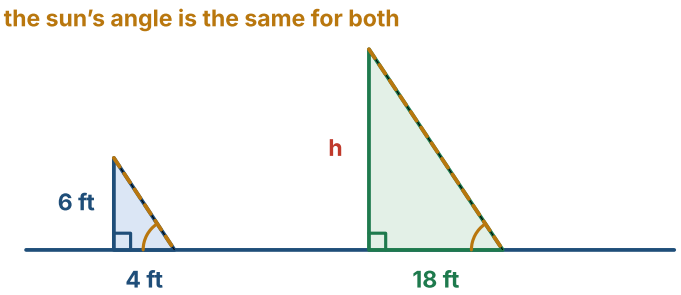
Solution.
AD/AB = 4/10.
k = **2.5** going from the small triangle to the large one.



5 Shadows, mirrors and indirect measurement

same sun angle → similar triangles → height/shadow is the same ratio for both

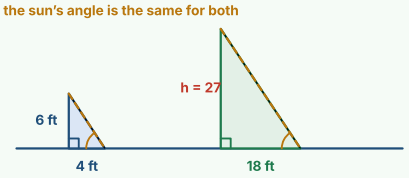
The classic SAT word problem. Write the ratio the same way round for both objects: height ÷ shadow for one equals height ÷ shadow for the other.



Both objects stand upright and both shadows run along the same ground, so the two triangles have the same three angles. 6/4 = h/18 gives h = 27.

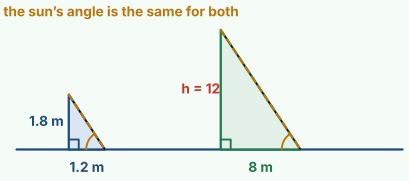
Example 1. **A 6-foot pole casts a 4-foot shadow. A signpost casts an 18-foot shadow. Find its height.**

Solution.
6/4 = h/18, so h = 18 × 1.5 = **27 feet**.



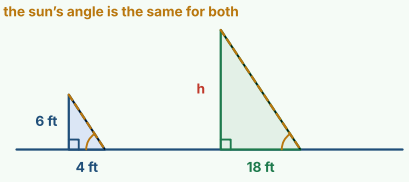
Example 2. **A 1.8 m person casts a 1.2 m shadow. A tree's shadow is 8 m. Find the tree's height.**

Solution.
1.8/1.2 = 1.5, and 8 × 1.5 = **12 m**.



Example 3. **Why must the two shadows be measured at the same time?**

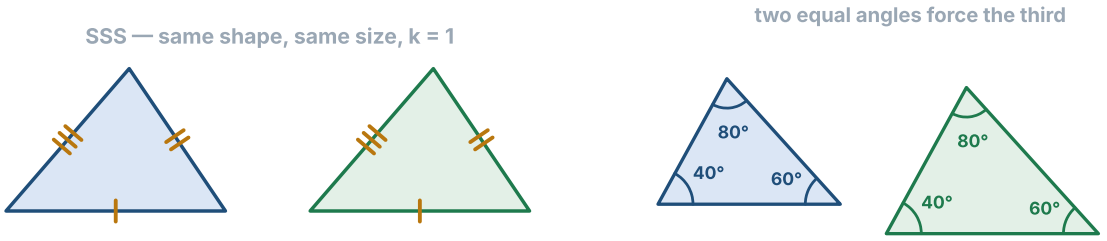
Solution.
The sun's angle changes through the day. Only at the same moment are the two triangles **similar**.



6 Congruence

congruent = same shape and same size, so the scale factor is 1

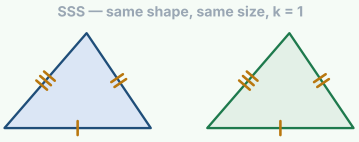
Congruent triangles match exactly: SSS, SAS, ASA or AAS. Note that AAA gives similarity only, never congruence, because the sizes can differ.



On the left the three pairs of sides match, so the triangles are congruent. On the right only the angles match, so they are similar — one is an enlargement of the other.

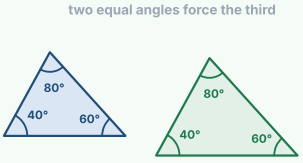
Example 1. **Two triangles have all three pairs of sides equal. Are they congruent?**

Solution.
Yes, by SSS.



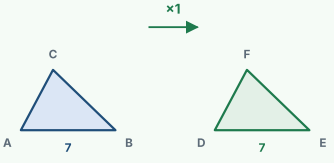
Example 2. **Two triangles have all three pairs of angles equal. Are they congruent?**

Solution.
Not necessarily — they are similar, but one could be an enlargement of the other.



Example 3. **Triangles ABC and DEF are congruent with AB = 7. What is DE?**

Solution.
Congruent means the scale factor is 1.
DE = **7**.



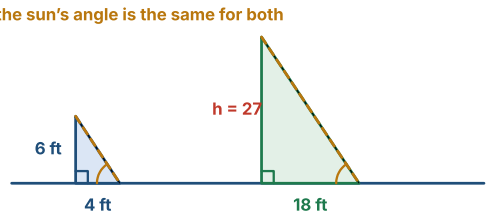
Part 1 — With Desmos (10 questions · draw the figure; the picture is faster than the algebra)

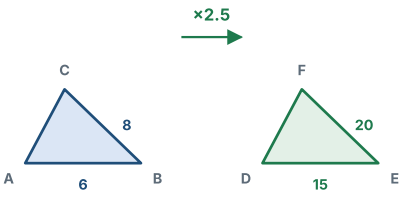
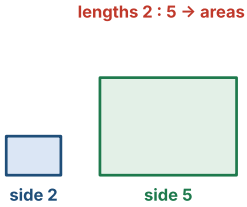
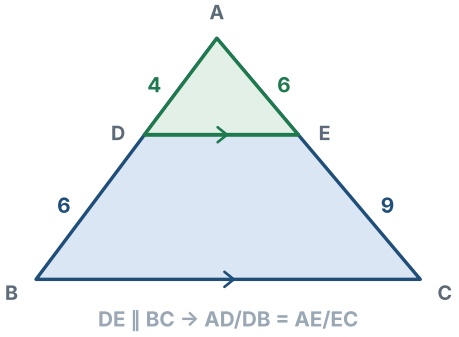
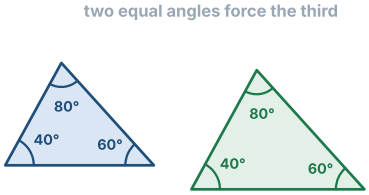
The SAT gives you Desmos on every maths question. Here it is **a proportion solver**. Type the ratio as an equation with the unknown in it and read the answer straight off.

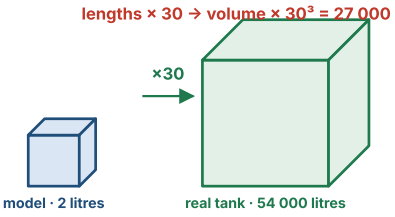
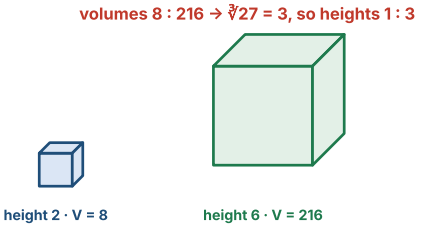
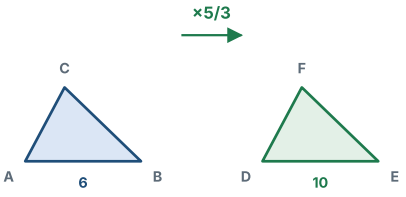
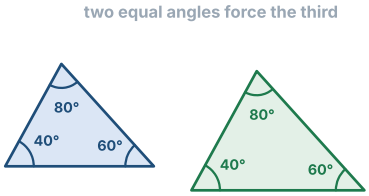
#	Question	Type into Desmos	Then	Answer
D1	A triangle with sides 3, 4, 5 is enlarged so the shortest side becomes 9. What is the new longest side?	`5*(9/3)`	Find the factor, then scale.	15
D2	Triangle ABC has AB = 6; similar triangle DEF has DE = 15. If BC = 8, what is EF?	`8*(15/6)`	The same factor applies to every side.	20
D3	Two similar rectangles have sides in the ratio 2 : 5. What is the ratio of their areas as a decimal?	`(2/5)^2`	Square the length ratio.	0.16
D4	Two similar solids have volumes 8 and 216. What is the ratio of their heights?	`(216/8)^(1/3)`	Cube root of the volume ratio.	3
D5	A model is built to scale 1 : 30 and holds 2 litres. How much does the real tank hold?	`2*30^3`	Volume scales by the cube.	54000
D6	A 6-foot pole casts a 4-foot shadow. A signpost casts an 18-foot shadow. How tall is it?	`18*(6/4)`	Same ratio for both objects.	27
D7	A 1.8 m person casts a 1.2 m shadow. A tree's shadow is 8 m. How tall is the tree?	`8*(1.8/1.2)`	Height over shadow is constant.	12
D8	In triangle ABC, DE is parallel to BC with AD = 4, DB = 6 and AE = 6. What is EC?	`6*(6/4)`	Set up AD/DB = AE/EC.	9
D9	Two similar triangles have areas 9 and 25. What is the ratio of their corresponding sides?	`sqrt(9/25)`	Take the square root of the area ratio.	0.6
D10	A photo 4 cm by 6 cm is enlarged by a factor of 2.5. What is the new area in square centimetres?	`4*6*2.5^2`	Area scales by the square of the factor.	150

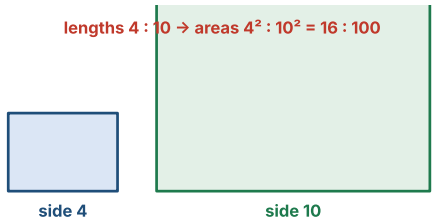
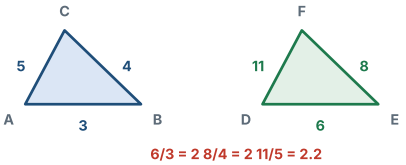
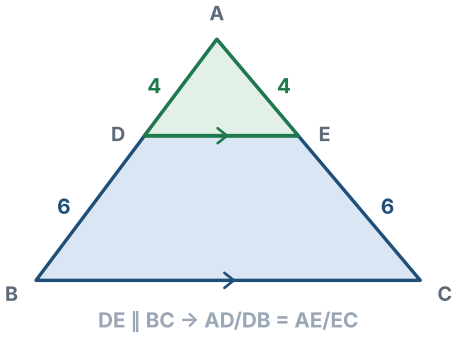
Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. For every question, first ask whether the quantity being scaled is a **length**, an **area** or a **volume**. That one decision picks the exponent.

#	Level	Question	Answer	Solution
P1	Hard	Beside a market stall, a vertical pole 6 feet tall casts a shadow 4 feet long. At the same time of day, a signpost casts a shadow 18 feet long. What is the height of the signpost, in feet? 	27	The sun's angle is the same, so the triangles are similar: 6/4 = h/18. h = 18 × 1.5 = **27 feet**. A common wrong answer comes from 18 − 6, not a ratio at all.

#	Level	Question	Answer	Solution
P2	Hard	Triangle ABC is similar to triangle DEF, with AB = 6 and DE = 15. If BC = 8, what is the length of EF? 	20	The scale factor is 15/6 = 2.5, and it applies to every pair of matching sides. EF = 8 × 2.5 = **20**.
P3	Hard	Two similar rectangles have corresponding sides in the ratio 2 : 5. What is the ratio of their areas? 	4 : 25	Area scales by the **square** of the length ratio. $2^2 : 5^2$ = **4 : 25**. A common wrong answer forgets to square; D cubes it, which is the volume rule for solids.
P4	Hard	In triangle ABC, point D lies on AB and point E lies on AC so that DE is parallel to BC. If AD = 4, DB = 6 and AE = 6, what is the length of EC? 	9	DE parallel to BC makes the two triangles similar, so AD/DB = AE/EC. 4/6 = 6/EC gives EC = 36/4 = **9**.
P5	Hard	Two triangles have angles of 40° and 60° in the first, and 40° and 80° in the second. State what is true here. 	They are similar, because two angles of each match.	The first triangle's third angle is 80° and the second's is 60°, so both have angles 40, 60 and 80. **Similar by AA.** Congruence would need the sizes to match too, which nothing here establishes.

#	Level	Question	Answer	Solution
P6	Hard	A model of a water tank is built to a scale of 1 : 30. The model holds 2 litres. How many litres does the real tank hold? 	54,000	Volume scales by the **cube** of the length factor: $30^3 = 27{,}000$. 2 × 27,000 = **54,000 litres**. A common wrong answer multiplies by 30 once; B squares it.
P7	Very hard	Two similar solid figures have volumes of 8 cubic centimetres and 216 cubic centimetres. What is the ratio of their corresponding heights? 	1 : 3	216/8 = 27, and the height ratio is the **cube root** of the volume ratio. $\sqrt[3]{27} = 3$, so the ratio is **1 : 3**. A common wrong answer uses the volume ratio itself.
P8	Very hard	Two similar triangles have areas of 9 and 25 square units. The shorter triangle has a side of length 6. What is the length of the corresponding side in the larger triangle? 	10	The side ratio is the square root of the area ratio: $\sqrt{(25/9)} = 5/3$. 6 × 5/3 = **10**.
P9	Very hard	Triangles ABC and DEF have all three pairs of angles equal. Which conclusion is guaranteed? 	The triangles are similar.	Equal angles guarantee the same **shape** but say nothing about size. **Similar**, and possibly congruent, but AAA alone never proves congruence — one could be an enlargement of the other.

#	Level	Question	Answer	Solution
P10	Very hard	A rectangular photograph measuring 4 cm by 6 cm is enlarged by a scale factor of 2.5. What is the area of the enlarged photograph, in square centimetres? 	150	Original area 24 cm²; area scales by 2.5² = 6.25. 24 × 6.25 = **150**. A common wrong answer multiplies the area by 2.5 once, the commonest error here.
P11	Very hard	Two triangles have sides 3, 4, 5 and 6, 8, 11 respectively. Are they similar, and why? 	No, because the three ratios are not all equal.	6/3 = 2 and 8/4 = 2, but 11/5 = 2.2. Similarity needs **all** matching ratios to agree, so they are not similar. A common wrong answer is true but irrelevant — similar triangles normally do have different perimeters.
P12	Very hard	In a figure, triangle ADE sits inside triangle ABC with DE parallel to BC. AD = 4, AB = 10 and the area of triangle ADE is 12 square units. What is the area of triangle ABC, in square units? 	75	The length factor is AB/AD = 10/4 = 2.5, so the area factor is 2.5² = 6.25. 12 × 6.25 = **75**. A common wrong answer applies the length factor 2.5 to the area directly instead of squaring it.

The one habit to build. Areas scale by k squared and volumes by k cubed. Your class scored 47% here, and almost every lost mark was a length factor applied to an area or a volume.

All 18 worked examples and all 22 answers checked numerically (see verify.py); every figure is drawn to the numbers in the question.

SAT Math 4.3 — Right Triangles & Trigonometry

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

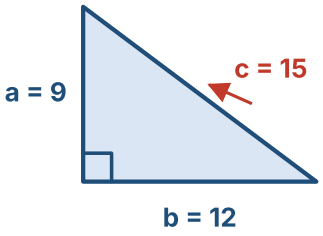
1 Pythagoras

$a^2 + b^2 = c^2$, where c is the hypotenuse — the side opposite the right angle

Identify the hypotenuse before substituting. It is always the longest side and always opposite the right angle. Putting a leg where c belongs is the error that costs this topic its marks.

$$a^2 + b^2 = c^2$$

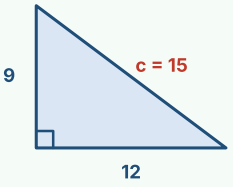
$$81 + 144 = 225$$



The arrow points from the right angle to the side facing it: that side is c. Here 81 + 144 = 225, so c = 15.

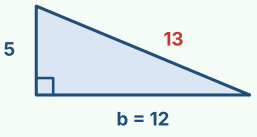
Example 1. **A right triangle has legs 9 and 12. Find the hypotenuse.**

Solution.
81 + 144 = 225.
c = √225 = **15**.



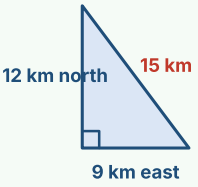
Example 2. **A right triangle has hypotenuse 13 and one leg 5. Find the other leg.**

Solution.
Here 13 is c, so 25 + b^2 = 169.
b = √144 = **12**.



Example 3. **A drone flies 9 km east then 12 km north. How far is it from the start?**

Solution.
The path makes a right angle.
√(81 + 144) = **15 km**.

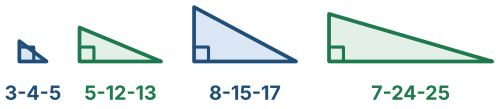


2 The triples worth memorising

3-4-5 | 5-12-13 | 8-15-17 | 7-24-25 — and all their multiples

Recognising a triple turns a calculation into a glance. 9-12-15 is 3-4-5 tripled; 10-24-26 is 5-12-13 doubled. If two sides fit a triple, the third is immediate.

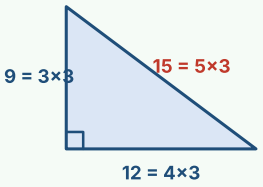
and every multiple: 6-8-10, 9-12-15, 10-24-26 ...



Four right triangles whose sides are whole numbers. Spotting one saves squaring, adding and taking a root.

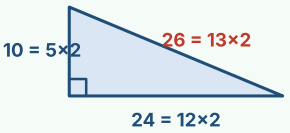
Example 1. **A right triangle has legs 9 and 12. Which triple is this?**

Solution.
3-4-5 multiplied by 3.
The hypotenuse is 5 × 3 = **15**.



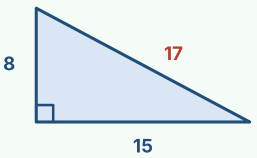
Example 2. **A right triangle has hypotenuse 26 and one leg 10. Find the other leg.**

Solution.
5-12-13 doubled.
The other leg is 12 × 2 = **24**.



Example 3. **Is a triangle with sides 8, 15, 17 right-angled?**

Solution.
64 + 225 = 289 = 17².
Yes — the converse of Pythagoras.

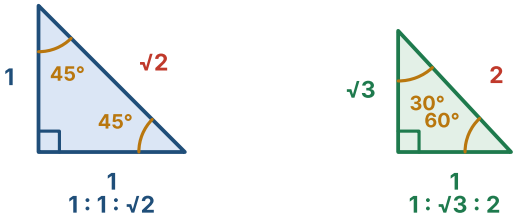


3 Special right triangles

45-45-90 → sides 1 : 1 : √2 | 30-60-90 → sides 1 : √3 : 2

These two appear constantly and are on the reference sheet. In a 30-60-90 the shortest side faces the 30° angle and is half the hypotenuse; the √3 side faces the 60°.

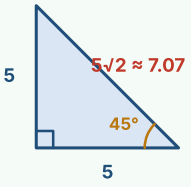
the short side faces 30° — half the hypotenuse



Memorise which side faces which angle. The hypotenuse is always the largest number in the ratio — √2 on the left, 2 on the right.

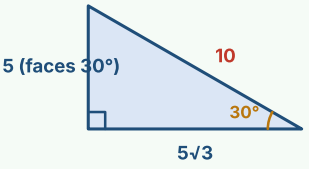
Example 1. **A 45-45-90 triangle has legs of 5. Find the hypotenuse.**

Solution.
5 × √2 = **5√2** ≈ 7.07.



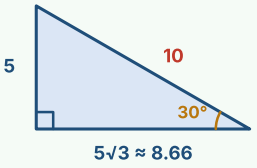
Example 2. **A 30-60-90 triangle has hypotenuse 10. Find the shortest side.**

Solution.
The side opposite 30° is half the hypotenuse.
5.



Example 3. **For that triangle, find the remaining side.**

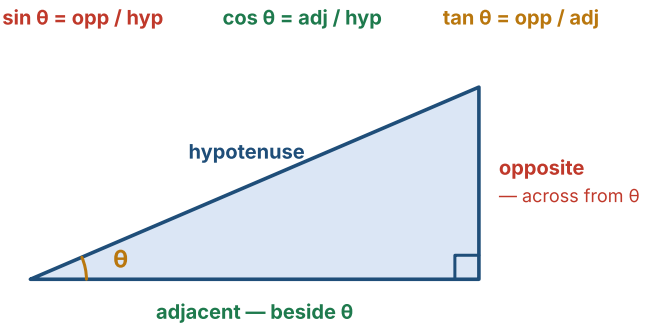
Solution.
It faces the 60° angle: 5√3 ≈ **8.66**.



4 SOH-CAH-TOA

sin = opp/hyp | cos = adj/hyp | tan = opp/adj

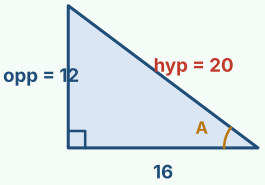
Label the three sides relative to the angle you are working with before choosing a ratio. Opposite and adjacent swap when you switch angles; the hypotenuse never moves.



Stand at θ: the side across the triangle is opposite, the side you are standing on is adjacent, and the long side facing the right angle is the hypotenuse.

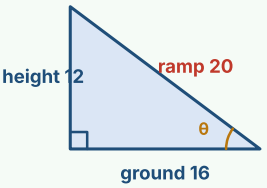
Example 1. **In a right triangle, sin A = 0.6 and the hypotenuse is 20. Find the side opposite A.**

Solution.
sin A = opp/hyp, so opp = 0.6 × 20 = **12**.



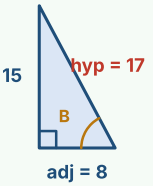
Example 2. **A ramp makes an angle whose sine is 0.6 and is 20 feet long. How high is the top?**

Solution.
The ramp is the hypotenuse.
0.6 × 20 = **12 feet**.



Example 3. **In a right triangle, the side adjacent to angle B is 8 and the hypotenuse is 17. Find cos B.**

Solution.
cos B = adj/hyp = **8/17** ≈ 0.471.

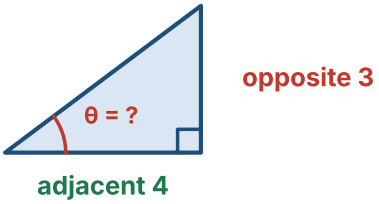


5 Finding an angle

use inverse trig: if $\tan\theta = 3/4$ then $\theta = \tan^{-1}(3/4)$

When the sides are known and the angle is not, the inverse functions give it. In Desmos type arctan, arcsin or arccos and make sure the calculator is in degrees.

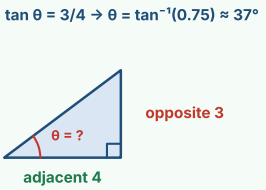
$$\tan\theta = 3/4 \rightarrow \theta = \tan^{-1}(0.75) \approx 37°$$



Two sides are known, the angle is not — so run the ratio backwards with the inverse function, and check the calculator says degrees.

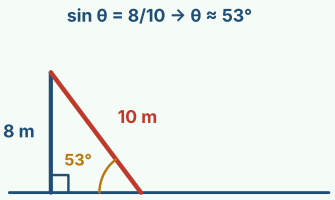
Example 1. **A right triangle has opposite 3 and adjacent 4. Find the angle, to the nearest degree.**

Solution.
$\tan\theta = 0.75$, so $\theta = \tan^{-1}(0.75) \approx$ **37°**.



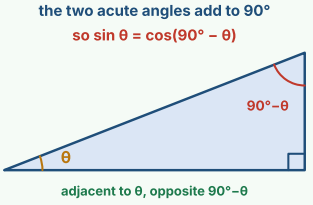
Example 2. **A ladder 10 m long reaches 8 m up a wall. What angle does it make with the ground?**

Solution.
$\sin\theta = 8/10 = 0.8$.
$\theta = \sin^{-1}(0.8) \approx$ **53°**.



Example 3. **Why do those two answers add to about 90°?**

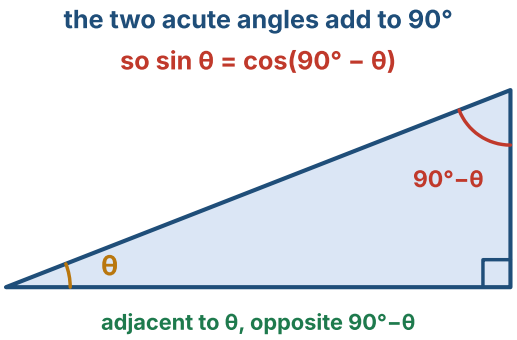
Solution.
They are the **two acute angles** of similar right triangles — 3-4-5 in both cases.



6 sin θ = cos(90° − θ)

the sine of an angle equals the cosine of its complement

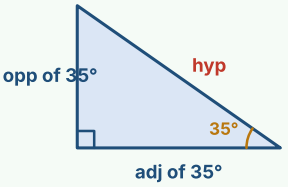
In a right triangle the two acute angles add to 90°, and what is opposite one is adjacent to the other. So sin and cos swap. Questions set sin(expression) = cos(expression) and ask you to make the two angles add to 90.



The side marked opposite for θ is the side adjacent for 90° − θ. Same two lengths, same fraction — only the name of the ratio changes.

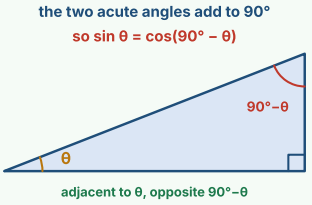
Example 1. **If sin 35° = cos x° and x is acute, find x.**

Solution.
x = 90 − 35 = **55**.



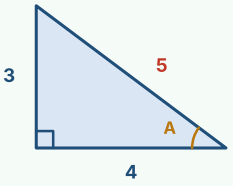
Example 2. **In a right triangle the acute angles are (2x + 10)° and (3x − 20)°. Find x.**

Solution.
They are complementary: 2x + 10 + 3x − 20 = 90, so 5x = 100.
x = **20**.



Example 3. **If sin A = 3/5 for an acute angle A, what is cos(90° − A)?**

Solution.
3/5 — it is the same quantity, by the identity.



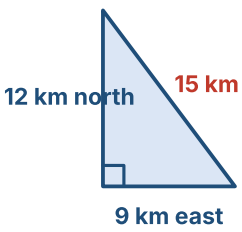
Part 1 — With Desmos (10 questions · draw the figure; the picture is faster than the algebra)

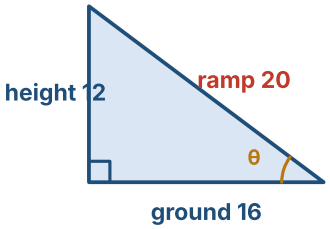
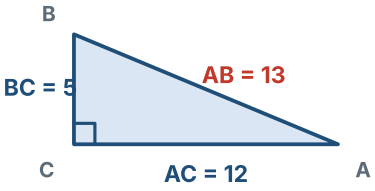
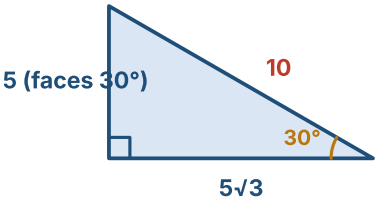
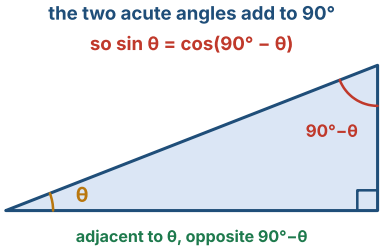
The SAT gives you Desmos on every maths question. Here it is **a trig calculator**. Use arcsin, arccos and arctan for angles, and check the setting is **degrees** before you trust anything.

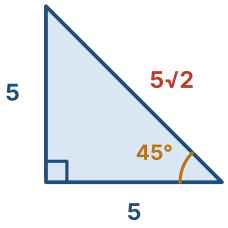
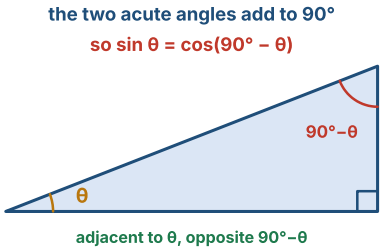
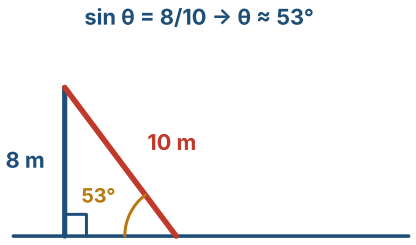
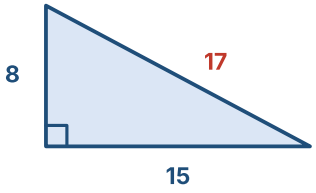
#	Question	Type into Desmos	Then	Answer
D1	A right triangle has legs 9 and 12. What is the hypotenuse?	`sqrt(9^2+12^2)`	Square, add, root — in one line.	15
D2	A right triangle has hypotenuse 13 and one leg 5. What is the other leg?	`sqrt(13^2-5^2)`	Subtract when you are given the hypotenuse.	12
D3	A 45-45-90 triangle has legs of 5. What is the hypotenuse, to two decimal places?	`5*sqrt(2)`	Desmos keeps the exact form too.	7.07
D4	A 30-60-90 triangle has hypotenuse 10. What is the side opposite 60°, to two decimals?	`5*sqrt(3)`	Half the hypotenuse, times √3.	8.66
D5	If sin A = 0.6 and the hypotenuse is 20, what is the opposite side?	`0.6*20`	sin = opp/hyp rearranged.	12
D6	A right triangle has opposite 3 and adjacent 4. What is the angle in degrees, to the nearest degree?	`arctan(3/4)`	Check Desmos is set to degrees.	37
D7	A ladder 10 m long reaches 8 m up a wall. What angle does it make with the ground?	`arcsin(8/10)`	Inverse sine.	53
D8	If sin 35° = cos x°, what is x?	`90-35`	Complementary angles.	55
D9	A right triangle has sides 8 and 17 as leg and hypotenuse. What is cos of that angle, to three decimals?	`8/17`	Adjacent over hypotenuse.	0.471
D10	In a right triangle the acute angles are (2x + 10)° and (3x − 20)°. What is x?	`2x+10+3x-20=90`	The two acute angles total 90.	20

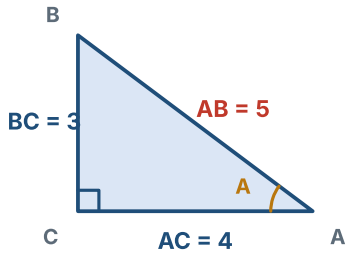
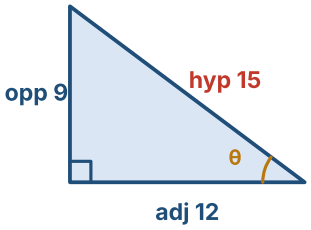
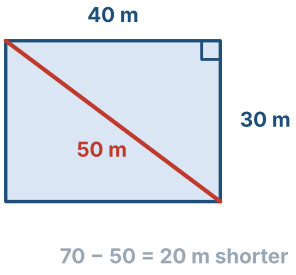
Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Label opposite, adjacent and hypotenuse relative to the angle in the question *before* choosing a ratio. Opposite and adjacent swap when the angle does.

#	Level	Question	Answer	Solution
P1	Hard	A drone takes off from a stadium, flies 9 kilometres due east, then 12 kilometres due north. What is the straight-line distance, in kilometres, between the drone and the stadium?  	15	East and north are perpendicular, so this is a right triangle. √(81 + 144) = √225 = **15 km**. It is 3-4-5 tripled.

#	Level	Question	Answer	Solution
P2	Hard	A straight loading ramp forms a right triangle with the level ground. The sine of the angle between the ramp and the ground is 0.6, and the ramp is 20 feet long. How many feet above the ground is the top of the ramp? 	12	The ramp is the hypotenuse, and sin = opp/hyp. 0.6 × 20 = **12 feet**. A common wrong answer divides 20 by 0.6 × 2 or similar — the clue is that the opposite side must be shorter than the hypotenuse.
P3	Hard	In right triangle ABC, the right angle is at C and the hypotenuse AB = 13. If BC = 5, what is the length of AC? 	12	AB is the hypotenuse, so $5^2 + AC^2 = 13^2$. AC = √144 = **12**. This is the 5-12-13 triple.
P4	Hard	A 30-60-90 triangle has a hypotenuse of length 10. What is the length of the side opposite the 30° angle? 	5	In a 30-60-90 triangle the side opposite 30° is **half the hypotenuse**. 10/2 = **5**. A common wrong answer is the side opposite 60°.
P5	Hard	If sin 35° = cos x° and 0 < x < 90, what is the value of x? 	55	sin θ = cos(90° − θ). x = 90 − 35 = **55**.

#	Level	Question	Answer	Solution
P6	Hard	A 45-45-90 triangle has legs of length 5. What is the length of the hypotenuse? 	5√2	In a 45-45-90 triangle the sides are in the ratio 1 : 1 : √2. The hypotenuse is **5√2** ≈ 7.07. A common wrong answer doubles the leg instead.
P7	Very hard	In a right-triangle design on a banner, the two acute angles measure (2x + 10)° and (3x − 20)°, so sin((2x + 10)°) = cos((3x − 20)°). What is the value of x? 	20	The identity holds exactly when the two angles are complementary. 2x + 10 + 3x − 20 = 90 gives 5x = 100, so x = **20**. The angles are 50° and 40°.
P8	Very hard	A ladder 10 metres long leans against a wall and reaches 8 metres up it. What angle, to the nearest degree, does the ladder make with the ground? 	53°	The wall height is opposite the angle and the ladder is the hypotenuse: sin θ = 0.8. θ = $\sin^{-1}(0.8)$ ≈ **53°**. A common wrong answer is the angle at the *top* of the ladder, the complement.
P9	Very hard	A right triangle has legs of length 8 and 15. What is the length of the hypotenuse? 	17	64 + 225 = 289, and √289 = **17** — the 8-15-17 triple. A common wrong answer comes from subtracting instead of adding the squares.

#	Level	Question	Answer	Solution
P10	Very hard	In right triangle ABC the right angle is at C, and sin A = 3/5. What is the value of cos B? 	3/5	Angles A and B are complementary, and sin A = cos(90° − A) = cos B. cos B = **3/5**. A common wrong answer is cos A, which answers a different question.
P11	Very hard	A right triangle has an angle whose tangent is 3/4. If the side adjacent to that angle is 12, what is the length of the hypotenuse? 	15	tan = opp/adj = 3/4, so with adj = 12 the opposite is 9. This is 9-12-15, which is 3-4-5 tripled, so the hypotenuse is **15**. A common wrong answer is the opposite side, not the hypotenuse.
P12	Very hard	A rectangular field measures 40 metres by 30 metres. A path runs diagonally from one corner to the opposite corner. How much shorter, in metres, is the diagonal path than walking along two sides? 	20	The diagonal is √(1600 + 900) = √2500 = 50 m, and two sides total 70 m. 70 − 50 = **20 metres shorter**.

The one habit to build. Find the hypotenuse first and write it down. Your class scored 20% on Pythagoras, and the usual cause is putting a leg where c belongs — the hypotenuse is always opposite the right angle and always the longest side.

All 18 worked examples and all 22 answers checked numerically (see verify.py); every figure is drawn to the numbers in the question.

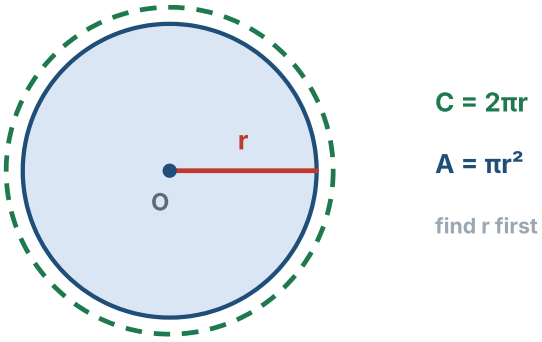
SAT Math 4.4 — Circles

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

1 Circumference and area

C = 2πr | A = πr² — both are on the reference sheet

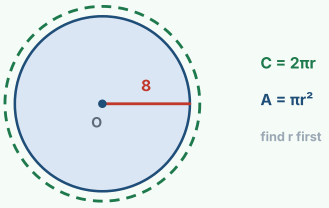
Almost every circle question starts by finding r. If you are given the area, take the square root; if the circumference, divide by 2π. Then answer what was asked.



The radius is the one number that unlocks both formulas: the green ring is the circumference and the shaded disc is the area.

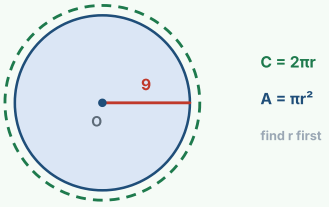
Example 1. **A circle has area 64π. Find its circumference.**

Solution.
$\pi r^2 = 64\pi$ gives $r = 8$.
$C = 2\pi(8) =$ **16π**.



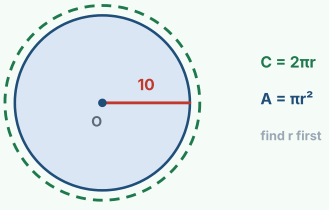
Example 2. **A circle has circumference 18π. Find its area.**

Solution.
$2\pi r = 18\pi$ gives $r = 9$.
$A = \pi(81) =$ **81π**.



Example 3. **A circular path encloses 100π square metres. How long is the path?**

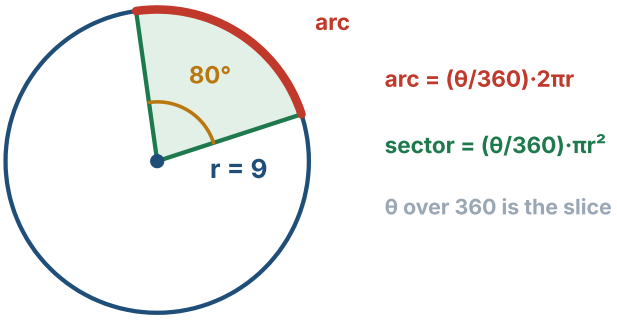
Solution.
$r = 10$, so the path is the circumference.
$2\pi(10) =$ **20π** metres.



2 Arcs and sectors

arc = (θ/360) × 2πr | sector area = (θ/360) × πr²

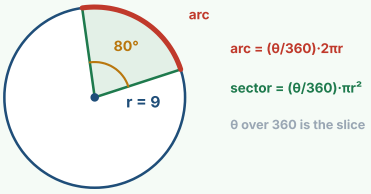
A sector is a fraction of the whole circle, and the fraction is the angle over 360. Work out the whole first, then take the slice.



The central angle picks the fraction: 80/360 = 2/9. Multiply the whole circumference by it for the arc, or the whole area for the sector.

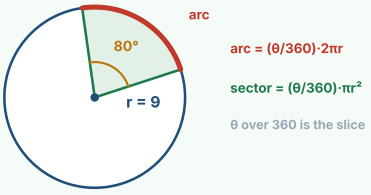
Example 1. **A circle of radius 9 has a sector with central angle 80°. Find its area.**

Solution.
Whole area = 81π, and 80/360 = 2/9.
(2/9)(81π) = **18π**.



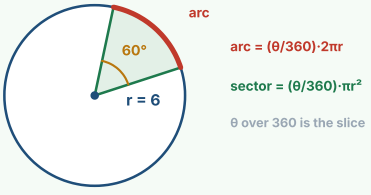
Example 2. **For that same circle, find the arc length of the sector.**

Solution.
Whole circumference = 18π.
(2/9)(18π) = **4π**.



Example 3. **A sector of a circle of radius 6 has area 6π. Find its central angle.**

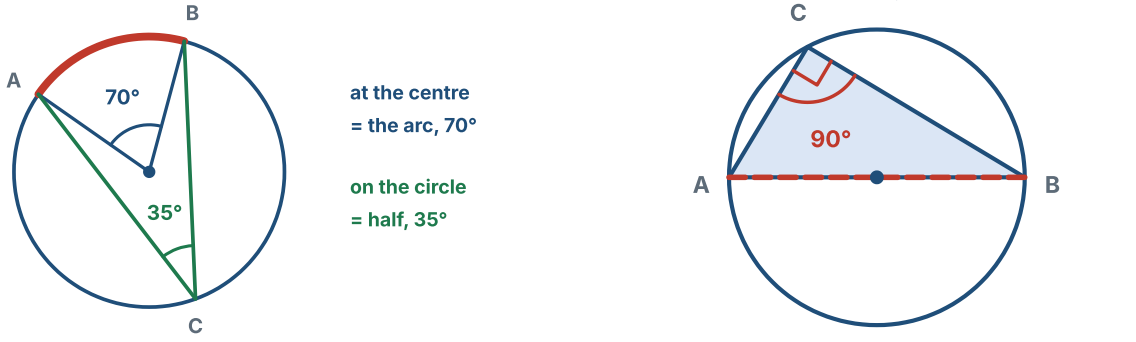
Solution.
Whole area 36π, and 6π/36π = 1/6.
(1/6)(360) = **60°**.



3 Central and inscribed angles

a central angle equals its arc | an inscribed angle is half its arc

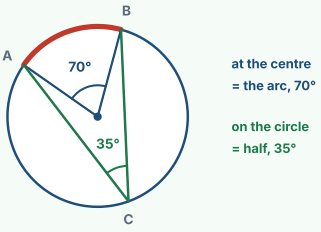
The vertex position decides the rule: at the centre, the angle equals the arc; on the circle, it is half. The special case worth knowing cold is that an angle inscribed in a semicircle is a right angle.



Both angles stand on the same red arc. The one at the centre is 70°; the one on the circle is 35°. When the arc is a semicircle (180°), half of it is a right angle.

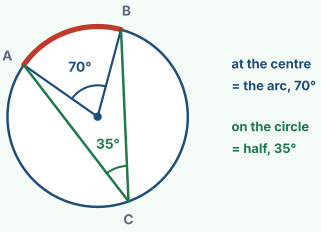
Example 1. **A central angle cuts an arc of 70°. What is the central angle?**

Solution.
70° — they are equal.



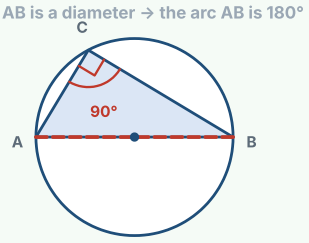
Example 2. **An inscribed angle cuts that same 70° arc. What is the inscribed angle?**

Solution.
Half the arc.
35°.



Example 3. **Triangle ABC is inscribed in a circle with AB a diameter. What is angle ACB?**

Solution.
The arc AB is 180°, and the inscribed angle is half.
90°.



4 Tangents and chords

a tangent is perpendicular to the radius at the point of contact

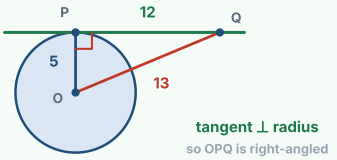
A tangent gives you a right angle for free, which usually means Pythagoras is next. A radius drawn perpendicular to a chord also bisects it.



Left: the radius OP meets the tangent at 90°, so OPQ is a right triangle with legs 5 and 12. Right: the perpendicular from the centre cuts the chord in half, giving legs 8 and d with hypotenuse 10.

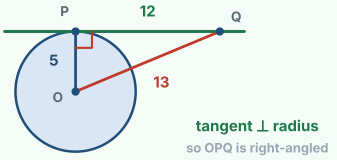
Example 1. **A tangent touches a circle of radius 5 at P. From an external point Q, PQ = 12. Find OQ.**

Solution.
Angle OPQ is 90°, so $OQ^2 = 25 + 144$.
OQ = **13**.



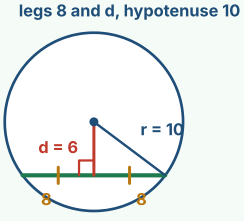
Example 2. **Why is triangle OPQ right-angled?**

Solution.
OP is a radius to the point of contact, and a **tangent is perpendicular to the radius** there.



Example 3. **A chord of length 16 sits in a circle of radius 10. How far is it from the centre?**

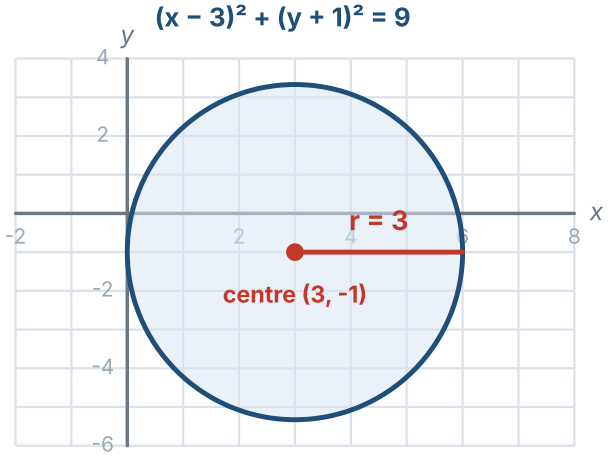
Solution.
The perpendicular from the centre bisects the chord, giving legs 8 and d with hypotenuse 10.
d = √(100 − 64) = **6**.



5 The equation of a circle

$(x - h)^2 + (y - k)^2 = r^2$, centre (h, k), radius r

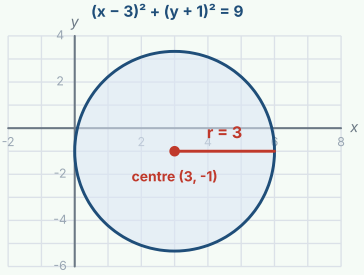
Read the centre with the signs flipped and remember the right-hand side is r squared, not r. This formula is *not* on the reference sheet.



$(x - 3)^2 + (y + 1)^2 = 9$ sits at $(3, -1)$: the x term reads -3 so $h = +3$, the y term reads $+1$ so $k = -1$. And 9 is r^2, so $r = 3$.

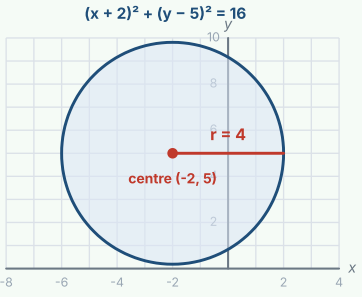
Example 1. **Find the centre and radius of $(x - 3)^2 + (y + 1)^2 = 9$.**

Solution.
Centre **$(3, -1)$**; radius **3**, because $9 = 3^2$.



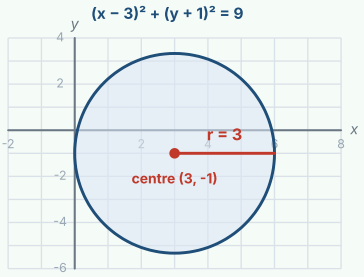
Example 2. **Write the equation of a circle with centre $(-2, 5)$ and radius 4.**

Solution.
$(x + 2)^2 + (y - 5)^2 = 16$.



Example 3. **Does the point $(6, -1)$ lie on the circle in the first example?**

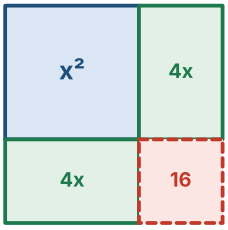
Solution.
$(6 - 3)^2 + (-1 + 1)^2 = 9 + 0 = 9$.
Yes, it is exactly on the circle.



6 Completing the square for a circle

group the x terms and the y terms, then halve and square each coefficient

An equation given as $x^2 + y^2 + Dx + Ey = F$ has to be rearranged before you can read anything off. Complete the square twice, once for x and once for y.



$x^2 + 8x$

$= (x + 4)^2 - 16$

halve 8, square it, then subtract it back

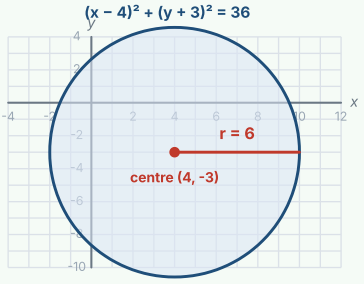
The square of side x plus the two strips of width 4 almost make a square of side x + 4. The dashed 16 is the missing corner — add it to both sides and the left-hand side becomes $(x + 4)^2$.

Example 1. **Find the radius of $x^2 + y^2 - 8x + 6y = 11$.**

Solution.

$(x - 4)^2 - 16 + (y + 3)^2 - 9 = 11$.

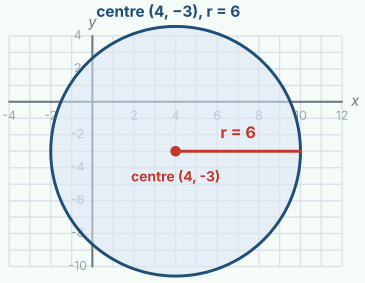
$(x - 4)^2 + (y + 3)^2 = 36$, so r = **6**.



Example 2. **Find its centre.**

Solution.

(4, −3) — the signs flip.

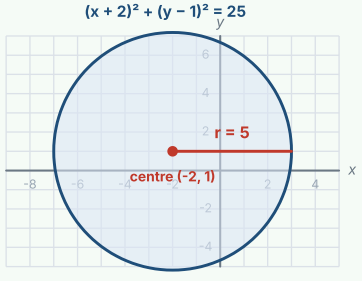


Example 3. **Find the radius of $x^2 + y^2 + 4x - 2y = 20$.**

Solution.

$(x + 2)^2 + (y - 1)^2 = 20 + 4 + 1 = 25$.

r = **5**.



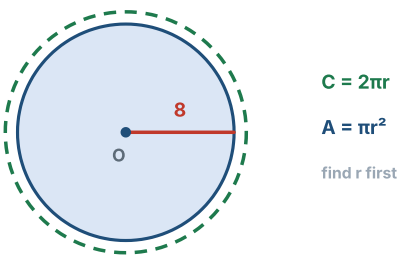
Part 1 — With Desmos (10 questions · draw the figure; the picture is faster than the algebra)

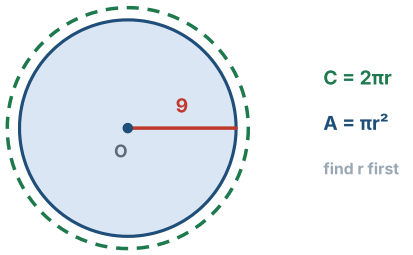
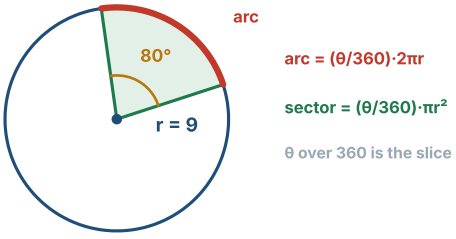
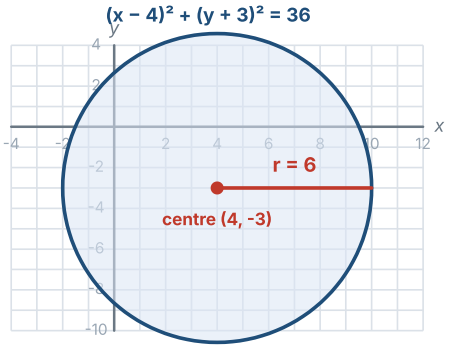
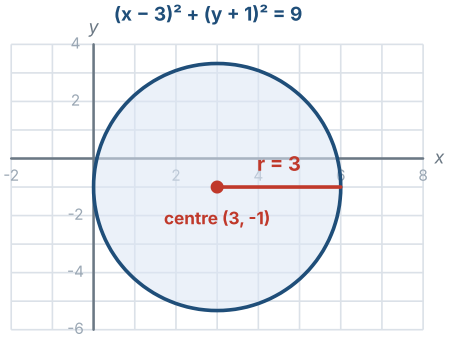
The SAT gives you Desmos on every maths question. Here it is **a circle grapher**. Type the equation in any form and Desmos draws it, centre and radius included — a fast check on your completing the square.

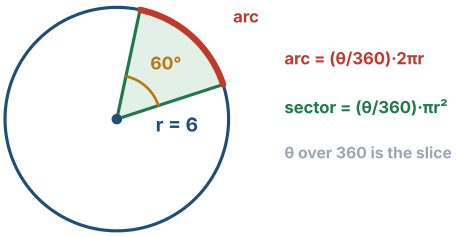
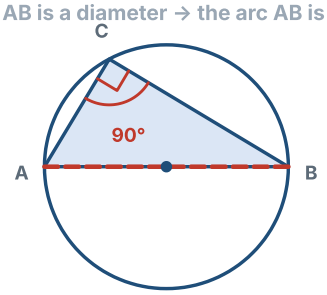
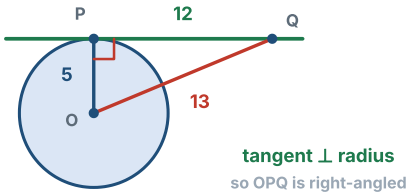
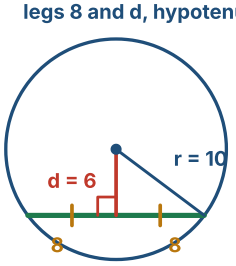
#	Question	Type into Desmos	Then	Answer
D1	A circle has area 64π. What is its circumference as a multiple of π?	`2*sqrt(64)`	Radius first, then the circumference.	16
D2	A circle has circumference 18π. What is its area as a multiple of π?	`(18/2)^2`	r = 9, then r squared.	81
D3	A sector of a circle of radius 9 has central angle 80°. What is its area as a multiple of π?	`80/360*81`	Fraction of the whole.	18
D4	For that same sector, what is the arc length as a multiple of π?	`80/360*18`	Same fraction, applied to the circumference.	4
D5	A sector of a circle of radius 6 has area 6π. What is its central angle in degrees?	`6/36*360`	The ratio of areas gives the ratio of angles.	60
D6	A tangent meets a circle of radius 5 at P, and PQ = 12. What is OQ?	`sqrt(5^2+12^2)`	The tangent makes a right angle with the radius.	13
D7	A chord of length 16 is in a circle of radius 10. How far is it from the centre?	`sqrt(10^2-8^2)`	Half the chord is one leg.	6
D8	What is the radius of $x^2 + y^2 - 8x + 6y = 11$?	`sqrt(11+16+9)`	Add the completing-the-square constants to both sides.	6
D9	What is the radius of $x^2 + y^2 + 4x - 2y = 20$?	`sqrt(20+4+1)`	Same method, different numbers.	5
D10	An inscribed angle cuts an arc of 70°. What is the inscribed angle?	`70/2`	Half the arc.	35

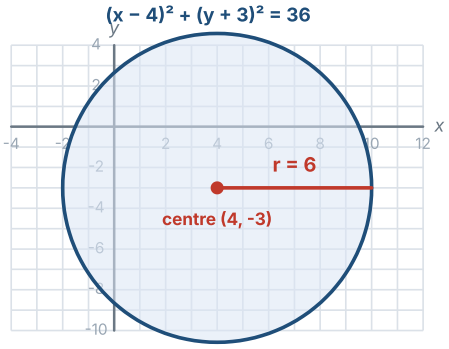
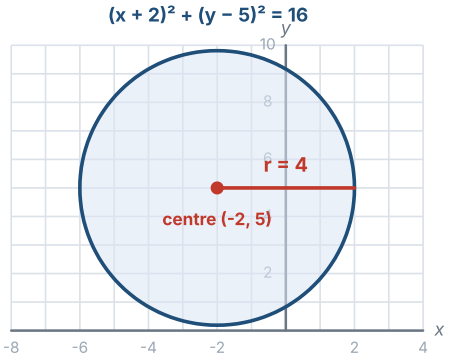
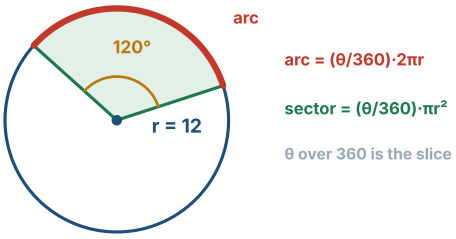
Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Find the radius first in almost every question, then answer what was actually asked: circumference, area, arc or sector.

#	Level	Question	Answer	Solution
P1	Hard	The area enclosed by a circular jogging path is 64π square metres. What is the length of the path, in metres? 	16π	$\pi r^2 = 64\pi$ gives r = 8, and the path is the circumference. $2\pi(8) =$ **16π**. A common wrong answer is the radius times π, a common slip.

#	Level	Question	Answer	Solution
P2	Hard	A round cheese wheel has a circumference of 18π centimetres. What is the area, in square centimetres, of its circular top face? 	81π	2πr = 18π gives r = 9. A = π(9)² = **81π**. A common wrong answer squares the diameter instead of the radius.
P3	Hard	A wedge cut from a circular cheese of radius 9 inches has a central angle of 80°. What is the area, in square inches, of the top face of the wedge? 	18π	The whole area is 81π and the wedge is 80/360 = 2/9 of it. (2/9)(81π) = **18π**.
P4	Hard	What is the radius of the circle given by $x^2 + y^2 - 8x + 6y = 11$? 	6	Complete the square twice: $(x - 4)^2 + (y + 3)^2 = 11 + 16 + 9 = 36$. r = √36 = **6**. Answering 36 gives r^2, not r.
P5	Hard	A circle in the xy-plane has equation $(x - 3)^2 + (y + 1)^2 = 9$. What are the coordinates of its centre? 	(3, −1)	The form is $(x - h)^2 + (y - k)^2 = r^2$, so h = 3 and k = −1. **(3, −1)** — the signs flip. A common wrong answer flips them the wrong way.

#	Level	Question	Answer	Solution
P6	Hard	A sector of a circle of radius 6 has an area of 6π. What is its central angle, in degrees? 	60	The whole circle has area 36π, and 6π/36π = 1/6. (1/6)(360) = **60°**.
P7	Very hard	In a circle with centre O, AB is a diameter and C is a point on the circle distinct from A and B. What is the measure of angle ACB, in degrees? 	90	Angle ACB is inscribed in a semicircle, so it cuts an arc of 180° and is half of it. **90°** wherever C sits — which is exactly why any other answer is wrong.
P8	Very hard	A tangent touches a circle with centre O and radius 5 at point P. An external point Q lies on the tangent with PQ = 12. What is the length of OQ? 	13	The tangent is perpendicular to the radius OP, so triangle OPQ has a right angle at P. OQ = √(25 + 144) = **13**, the 5-12-13 triple.
P9	Very hard	A chord of length 16 is drawn in a circle of radius 10. What is the distance from the centre of the circle to the chord? 	6	The perpendicular from the centre bisects the chord, giving a right triangle with hypotenuse 10 and one leg 8. d = √(100 − 64) = **6**. A common wrong answer is half the chord itself.

#	Level	Question	Answer	Solution
P10	Very hard	The circular fountain plaza at the centre of a market square is modelled by $x^2 + y^2 - 8x + 6y = 11$, with units in metres. What is the radius, in metres? 	6	Completing the square gives $(x - 4)^2 + (y + 3)^2 = 36$. r = **6**. A common wrong answer is r^2, and C is the number on the right of the original equation — both are answers to steps along the way, not the question.
P11	Very hard	A circle has centre (−2, 5) and radius 4. Write the equation of this circle. 	$(x + 2)^2 + (y - 5)^2 = 16$	Signs flip inside the brackets and the right-hand side is r^2. **$(x + 2)^2 + (y - 5)^2 = 16$.** A common wrong answer uses r instead of r^2; another gets both signs wrong.
P12	Very hard	A circle of radius 12 centimetres has a sector whose arc length is 8π centimetres. What is the area of that sector, in square centimetres? 	48π	The full circumference is 24π, so the sector is 8π/24π = 1/3 of the circle. The full area is 144π, and (1/3)(144π) = **48π**.

The one habit to build. The right-hand side of a circle equation is r^2, not r. Your class scored 47% on the circle equation, and answering 36 instead of 6 was the single most common slip.

All 18 worked examples and all 22 answers checked numerically (see verify.py); every figure is drawn to the numbers in the question.

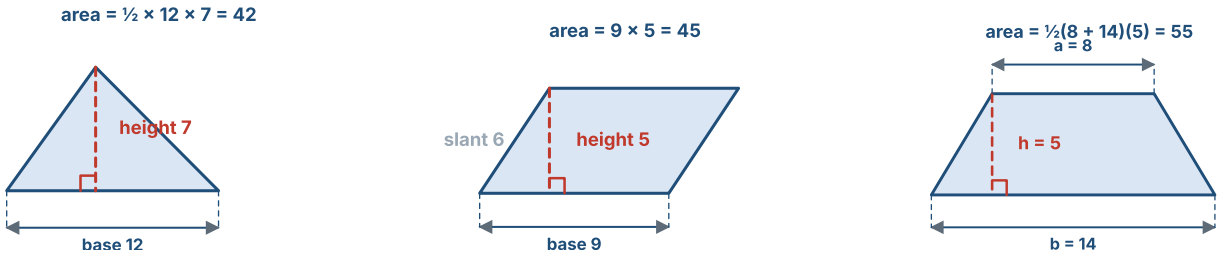
SAT Math 4.5 — Area & Volume

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

1 Areas of plane figures

triangle ½bh | parallelogram bh | trapezoid ½(a + b)h | circle πr²

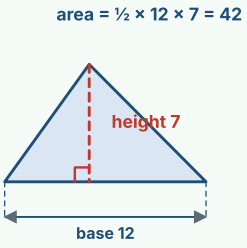
The height is always the *perpendicular* distance, not the slanted side. A triangle drawn leaning over still uses the vertical height.



In all three the red dashed line is the height: it meets the base at a right angle. The slanted side of 6 in the parallelogram is a distraction — the height is 5.

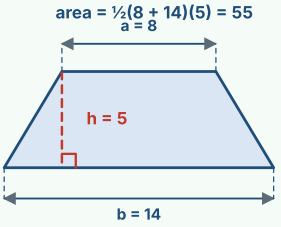
Example 1. **A triangle has base 12 and perpendicular height 7. Find its area.**

Solution.
½(12)(7) = **42**.



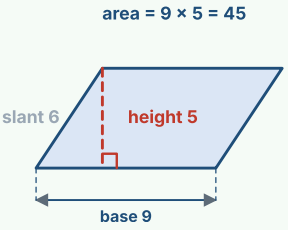
Example 2. **A trapezoid has parallel sides 8 and 14 and height 5. Find its area.**

Solution.
½(8 + 14)(5) = ½(22)(5) = **55**.



Example 3. **A parallelogram has base 9 and slant side 6, with perpendicular height 5. Find its area.**

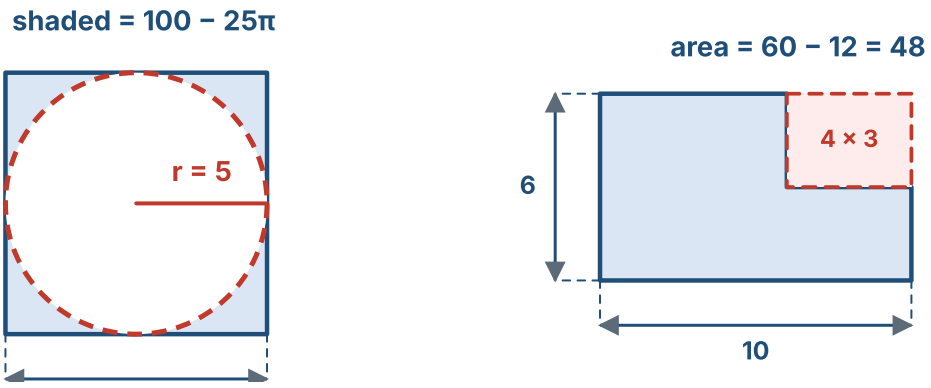
Solution.
Use the perpendicular height, not the slant.
9 × 5 = **45**.



2 Composite figures

split the shape into pieces you know, or subtract a hole from a whole

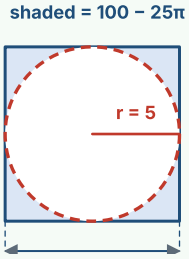
Decide whether adding or subtracting is quicker, then label every piece before calculating. A shaded region is nearly always one area minus another.



Left: the shaded part is the square minus the circular hole. Right: the L-shape is the full rectangle minus the dashed corner that was removed.

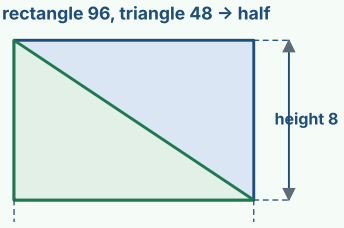
Example 1. **A square of side 10 has a circle of radius 5 cut from it. Find the shaded area left over.**

Solution.
100 − 25π ≈ 100 − 78.54 = **21.46**.



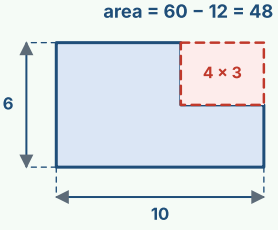
Example 2. **A rectangle 12 by 8 has a triangle of base 12 and height 8 removed. What fraction remains?**

Solution.
Rectangle 96, triangle 48.
Half remains.



Example 3. **An L-shape is a 10 by 6 rectangle with a 4 by 3 rectangle removed from one corner. Find its area.**

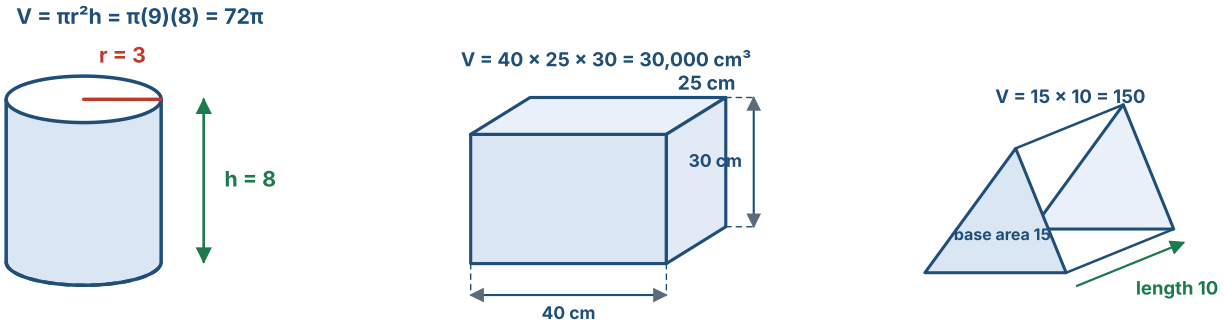
Solution.
60 − 12 = **48**.



3 Volume of prisms and cylinders

V = (area of the base) × height | cylinder V = πr²h

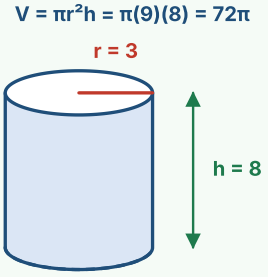
Any solid with the same cross-section all the way up follows one rule: find the base area, multiply by the height. The reference sheet gives these, so the work is choosing the base.



Each solid has one cross-section repeated along its length. Find that face's area — a circle, a rectangle, a triangle — then multiply by how far it travels.

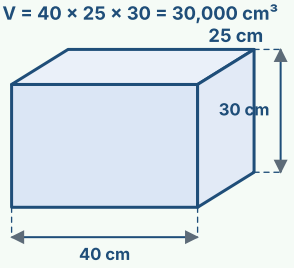
Example 1. **A cylinder has radius 3 and height 8. Find its volume.**

Solution.
π(9)(8) = **72π** ≈ 226.19.



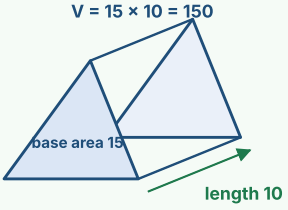
Example 2. **A rectangular tank measures 40 by 25 by 30 centimetres. Find its volume.**

Solution.
40 × 25 × 30 = **30,000 cm³**, which is 30 litres.



Example 3. **A triangular prism has a base triangle of area 15 and length 10. Find its volume.**

Solution.
15 × 10 = **150**.

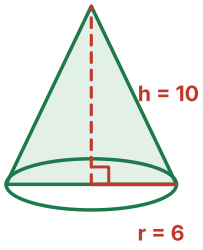


4 Cones, pyramids and spheres

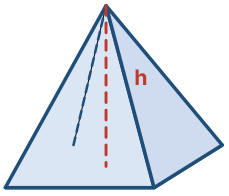
cone $\frac{1}{3}\pi r^2 h$ | pyramid $\frac{1}{3}Bh$ | sphere $\frac{4}{3}\pi r^3$

The one-third is the whole difficulty: a cone is a third of the cylinder that would contain it. All three formulas are on the reference sheet, so check you have copied the fraction.

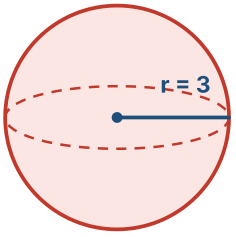
$V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(36)(10) = 120\pi$



$V = \frac{1}{3} \times \text{base area} \times h$



$V = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(27) = 36\pi$

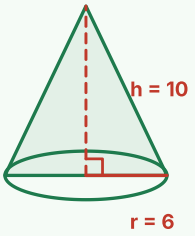


The cone and the pyramid both carry the one third; the sphere carries four thirds and cubes the radius. Copy the fraction before you substitute.

Example 1. **A cone has radius 6 and height 10. Find its volume.**

Solution.
(1/3)π(36)(10) = **120π**.

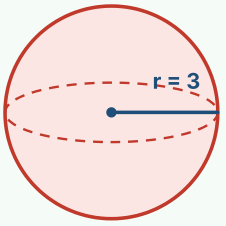
$V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(36)(10) = 120\pi$



Example 2. **A sphere has radius 3. Find its volume.**

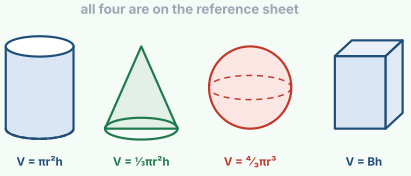
Solution.
(4/3)π(27) = **36π**.

$V = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(27) = 36\pi$



Example 3. **A cylinder and a cone share the same radius and height. How do their volumes compare?**

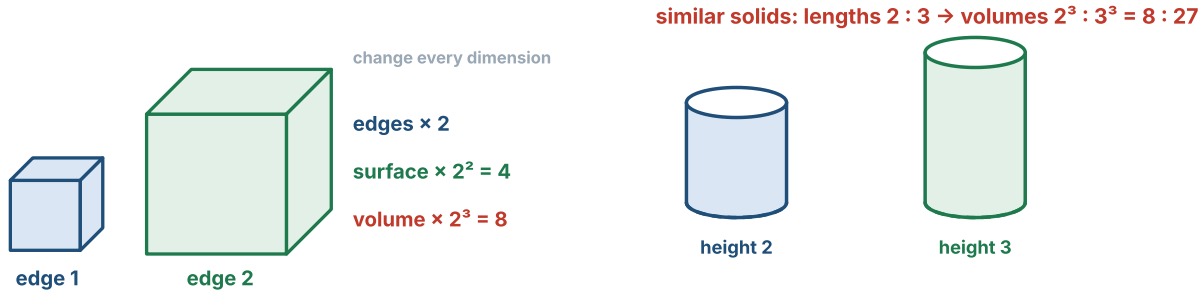
Solution.
The cone is exactly **one third** of the cylinder.



5 Scaling a solid

lengths × k | surface area × k^2 | volume × k^3

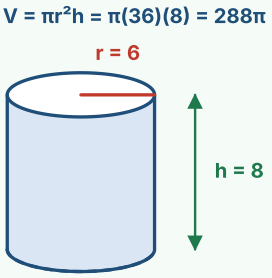
Changing one dimension multiplies the volume by that factor alone. Changing every dimension by k multiplies the volume by k cubed. Read carefully which the question means.



Left: every edge doubles, so the volume is eight times as large. Right: two similar cans in the ratio 2 : 3 have volumes in the ratio 8 : 27.

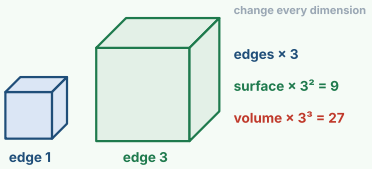
Example 1. **A cylinder's radius is doubled and its height is unchanged. What happens to the volume?**

Solution.
V depends on r^2.
The volume is multiplied by **4**.



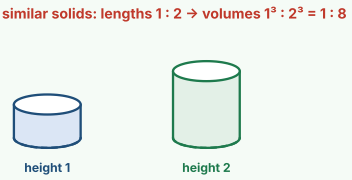
Example 2. **A cube's edges are all tripled. What happens to its volume?**

Solution.
Every dimension triples.
Volume × 3^3 = **27**.



Example 3. **A cylinder's height alone is doubled. What happens to the volume?**

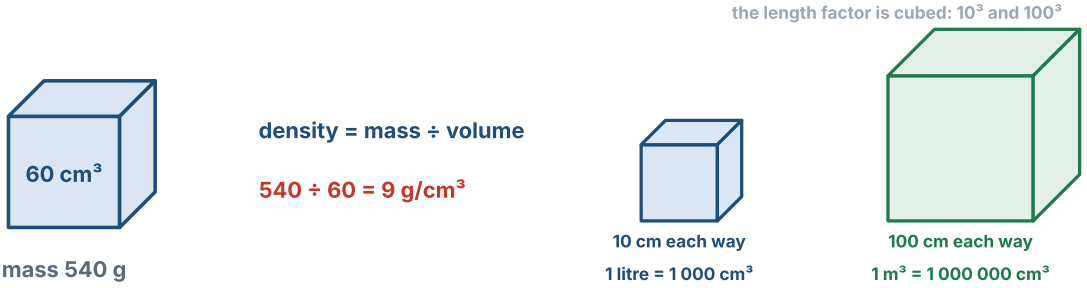
Solution.
Only one dimension changes.
Volume **doubles**.



6 Density and unit conversion

density = mass ÷ volume | for volume the conversion factor is cubed

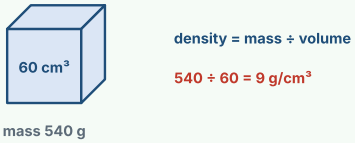
Density questions are a rate in disguise. For units, remember 1 litre = 1000 cm^3 while 1 m^3 = 1,000,000 cm^3 — the two are not the same factor.



A litre is a 10 cm cube, so 10^3 = 1 000 cm^3. A cubic metre is a 100 cm cube, so 100^3 = 1 000 000 cm^3. Using one factor for the other is the classic slip.

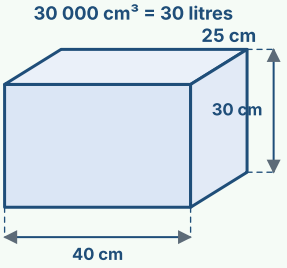
Example 1. **A block has mass 540 g and volume 60 cm^3. Find its density.**

Solution.
540/60 = **9 g/cm^3**.



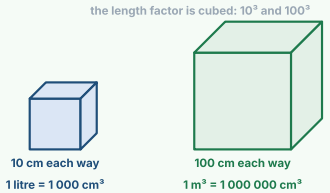
Example 2. **A tank holds 30,000 cm^3. How many litres is that?**

Solution.
30,000 ÷ 1000 = **30 litres**.



Example 3. **How many cubic centimetres are in 1 cubic metre?**

Solution.
100^3 = **1,000,000**. The length factor 100 gets cubed.



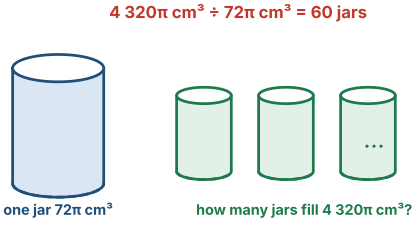
Part 1 — With Desmos (10 questions · draw the figure; the picture is faster than the algebra)

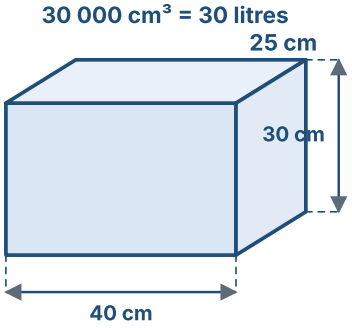
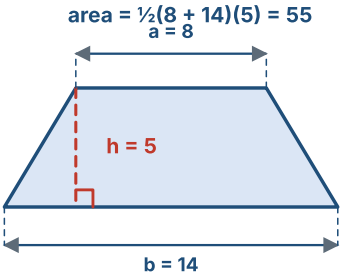
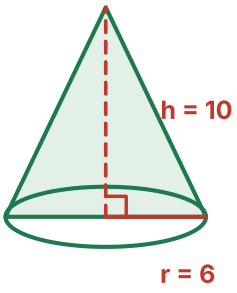
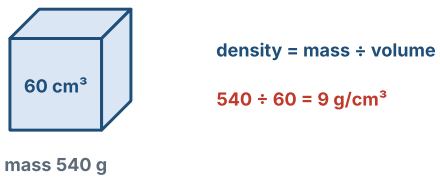
The SAT gives you Desmos on every maths question. Here it is **an arithmetic engine for formulas**. Type the whole formula with its numbers in one line so the 1/3 or the ½ cannot go missing.

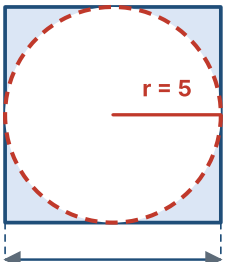
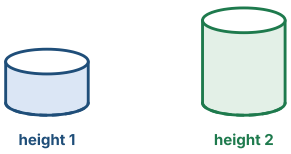
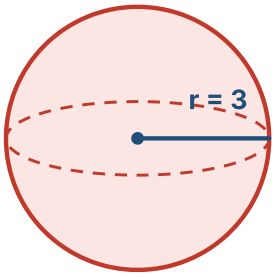
#	Question	Type into Desmos	Then	Answer
D1	A triangle has base 12 and height 7. What is its area?	`0.5*12*7`	Half base times height.	42
D2	A trapezoid has parallel sides 8 and 14 and height 5. What is its area?	`0.5*(8+14)*5`	Average the parallel sides, then multiply by the height.	55
D3	A cylinder has radius 3 and height 8. What is its volume as a multiple of π?	`3^2*8`	r squared times h.	72
D4	A cone has radius 6 and height 10. What is its volume as a multiple of π?	`(1/3)*6^2*10`	Do not forget the one third.	120
D5	A sphere has radius 3. What is its volume as a multiple of π?	`(4/3)*3^3`	Four thirds times r cubed.	36
D6	A tank measures 40 by 25 by 30 cm. How many litres does it hold?	`40*25*30/1000`	Divide by 1000, not by a million.	30
D7	A block has mass 540 g and volume 60 cm³. What is its density in g/cm³?	`540/60`	Mass over volume.	9
D8	A square of side 10 has a circle of radius 5 removed. What area remains, to two decimals?	`100-pi*25`	Subtract the hole from the whole.	21.46
D9	A cylinder's radius is doubled and its height is unchanged. By what factor does the volume grow?	`2^2`	Volume depends on r squared.	4
D10	How many cubic centimetres are in one cubic metre?	`100^3`	The length factor is cubed.	1000000

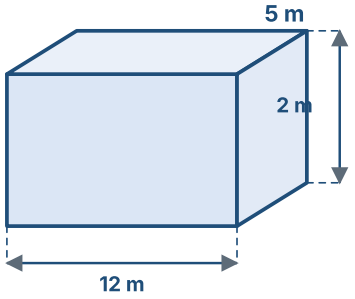
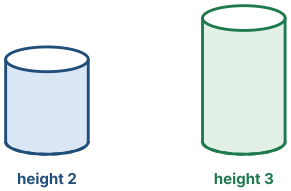
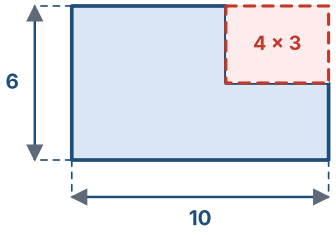
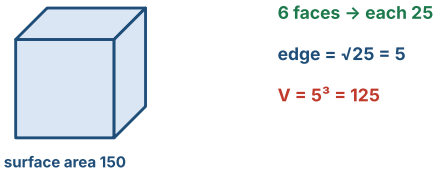
Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Write the formula down before substituting. Most lost marks in this topic are a correct calculation of the wrong formula.

#	Level	Question	Answer	Solution
P1	Hard	A market supplier pours 4,320π cubic centimetres of olive oil into identical cylindrical jars. Each jar has an inside radius of 3 centimetres and is filled to a height of 8 centimetres. How many jars does the supplier fill? 	60	Each jar holds π(9)(8) = 72π cm³. 4320π ÷ 72π = **60 jars**. A common wrong answer is the volume of one jar in multiples of π, not a count.

#	Level	Question	Answer	Solution
P2	Hard	An oil tin is a rectangular prism measuring 40 cm by 25 cm by 30 cm and is completely full. How many litres of oil does it hold? (1 litre = 1000 cubic centimetres) 	30	40 × 25 × 30 = 30,000 cm³. 30,000 ÷ 1000 = **30 litres**. Dividing by 1,000,000 gives 0.03 — that factor converts to cubic *metres*, and eight of your class used it on Test 5.
P3	Hard	A trapezoid has parallel sides of 8 and 14 and a perpendicular height of 5. What is its area? 	55	Area = ½(a + b)h = ½(22)(5) = **55**. A common wrong answer forgets the one half; another averages the sides but omits the height.
P4	Hard	A cone has a radius of 6 and a height of 10. What is its volume? $V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(36)(10) = 120\pi$ 	120π	V = (1/3)πr²h = (1/3)π(36)(10). = **120π**. A common wrong answer is the cylinder's volume — the one third was dropped.
P5	Hard	A metal block has a mass of 540 grams and a volume of 60 cubic centimetres. What is its density, in grams per cubic centimetre? 	9	Density = mass ÷ volume = 540/60 = **9 g/cm³**. A common wrong answer inverts the fraction; another multiplies instead of dividing.

#	Level	Question	Answer	Solution
P6	Hard	A square of side 10 centimetres has a circle of radius 5 centimetres cut out of it. What is the area of the remaining region, to the nearest hundredth of a square centimetre? **shaded = 100 − 25π** 	21.46	100 − π(25) = 100 − 78.5398… = **21.46 cm²**.
P7	Very hard	A cylinder's radius is doubled while its height stays the same. By what factor is its volume multiplied? similar solids: lengths 1 : 2 → volumes $1^3 : 2^3 = 1 : 8$ 	4	V = πr²h, and only r changes, so the factor is 2^2. The volume is **4 times** as large. A common wrong answer would be right if *every* dimension doubled.
P8	Very hard	A sphere and a cylinder both have radius 3. The cylinder has height 3. What is the ratio of the sphere's volume to the cylinder's? $V = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(27) = 36\pi$ 	4 : 3	Sphere = (4/3)π(27) = 36π; cylinder = π(9)(3) = 27π. 36 : 27 = **4 : 3**. A common wrong answer inverts the ratio.

#	Level	Question	Answer	Solution
P9	Very hard	A rectangular swimming pool is 12 metres long, 5 metres wide and 2 metres deep. Water is pumped in at 40 litres per minute. How many minutes does it take to fill? 120 m³ = 120 000 litres  	3000	Volume = 120 m³, and 1 m³ = 1000 litres, so 120,000 litres. 120,000 ÷ 40 = **3,000 minutes**. Using 1 m³ = 1,000,000 cm³ and then forgetting to convert to litres is the usual slip here.
P10	Very hard	Two similar cylindrical cans have heights in the ratio 2 : 3. What is the ratio of their volumes? similar solids: lengths 2 : 3 → volumes 2³ : 3³ = 8 : 27  	8 : 27	Similar solids scale by k in length, so volumes scale by k³. 2³ : 3³ = **8 : 27**. A common wrong answer squares instead of cubing, which is the surface-area rule.
P11	Very hard	An L-shaped floor is formed by removing a 4 m by 3 m rectangle from one corner of a 10 m by 6 m rectangle. Tiles cost $12 per square metre. What is the total cost of tiling the floor? area = 60 − 12 = 48  	$576	Area = 60 − 12 = 48 m². 48 × 12 = **$576**. A common wrong answer tiles only the 12 m² corner that was removed.
P12	Very hard	A cube has a surface area of 150 square centimetres. What is its volume, in cubic centimetres?  	125	A cube has 6 faces, so each face has area 150/6 = 25 and the edge is 5. V = 5³ = **125**. A common wrong answer is the area of one face, not the volume.

The one habit to build. Litres and cubic metres use different conversion factors: 1 litre = 1,000 cm³ but 1 m³ = 1,000,000 cm³. Eight of thirteen of your class used the wrong one on Test 5 and turned 30 litres into 0.03.

All 18 worked examples and all 22 answers checked numerically (see verify.py); every figure is drawn to the numbers in the question.

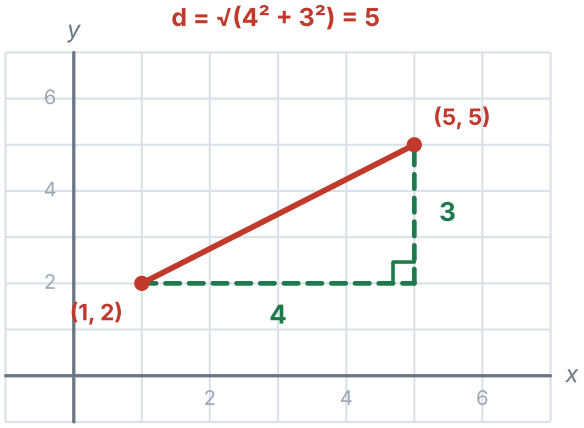
SAT Math 4.6 — Coordinate Geometry

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

1 Distance between two points

$$d = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]} \text{ — it is Pythagoras in disguise}$$

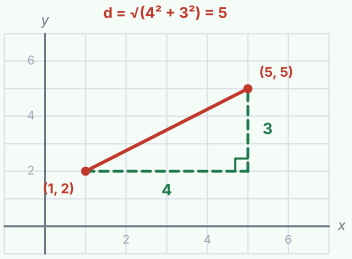
Draw the horizontal and vertical legs and the formula becomes obvious. Because both differences are squared, the order of subtraction does not matter.



The green legs are just the differences in x and in y: 4 across and 3 up. The red segment is the hypotenuse of that right triangle, so d = 5.

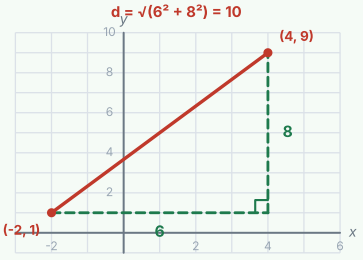
Example 1. **Find the distance between (1, 2) and (5, 5).**

Solution.
Legs 4 and 3.
√(16 + 9) = **5**.



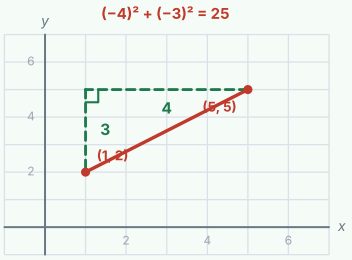
Example 2. **Find the distance between (−2, 1) and (4, 9).**

Solution.
Legs 6 and 8.
√(36 + 64) = **10**.



Example 3. **Why does subtracting the other way round give the same answer?**

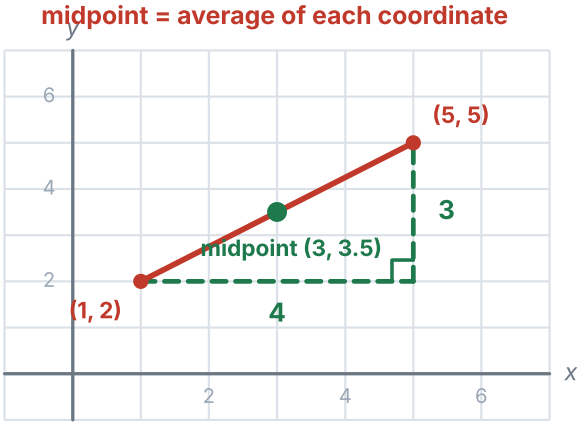
Solution.
Squaring removes the sign: $(-4)^2 = 4^2$.



2 Midpoint

midpoint = $((x_1 + x_2)/2, (y_1 + y_2)/2)$ — average each coordinate

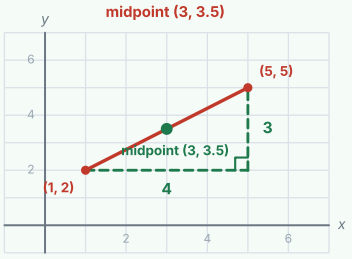
Average the x's and average the y's, separately. If you are given one endpoint and the midpoint, work backwards: the midpoint is as far past itself again.



The midpoint sits halfway along in both directions: x goes 1 → 3 → 5 and y goes 2 → 3.5 → 5.

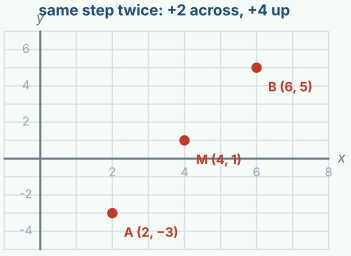
Example 1. **Find the midpoint of (1, 2) and (5, 5).**

Solution.
((1+5)/2, (2+5)/2) = **(3, 3.5)**.



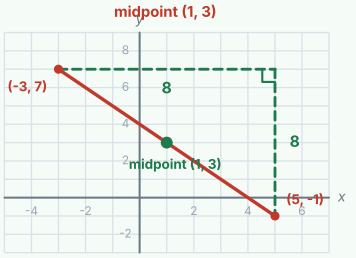
Example 2. **The midpoint of AB is (4, 1) and A is (2, −3). Find B.**

Solution.
B is the same step again: (4 + 2, 1 + 4) = **(6, 5)**.



Example 3. **Find the midpoint of (−3, 7) and (5, −1).**

Solution.
((−3+5)/2, (7−1)/2) = **(1, 3)**.

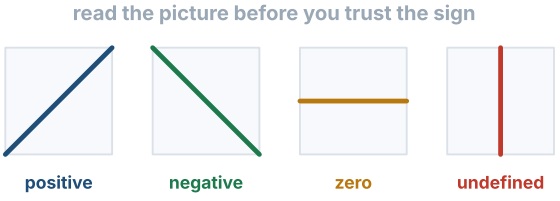


3 Slope

$$m = (y_2 - y_1)/(x_2 - x_1) \text{ — rise over run}$$

Keep the two points in the same order on top and bottom. A positive slope rises left to right; a horizontal line has slope 0 and a vertical line has no slope at all.

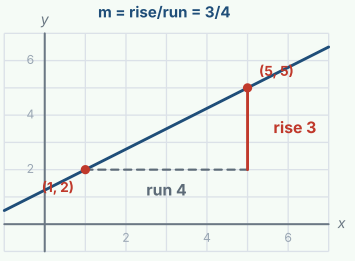




Rise 3 over run 4 gives m = 3/4. The four small panels show the only four possibilities: up, down, flat (0) and vertical (undefined).

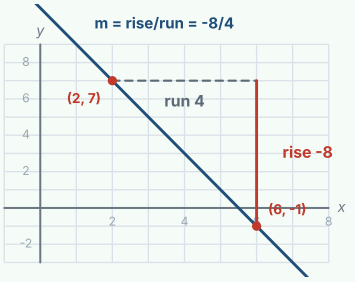
Example 1. **Find the slope through (1, 2) and (5, 5).**

Solution.
(5 − 2)/(5 − 1) = **3/4**.



Example 2. **Find the slope through (2, 7) and (6, −1).**

Solution.
(−1 − 7)/(6 − 2) = −8/4 = **−2**.



Example 3. **What is the slope of the line through (3, 4) and (3, 9)?**

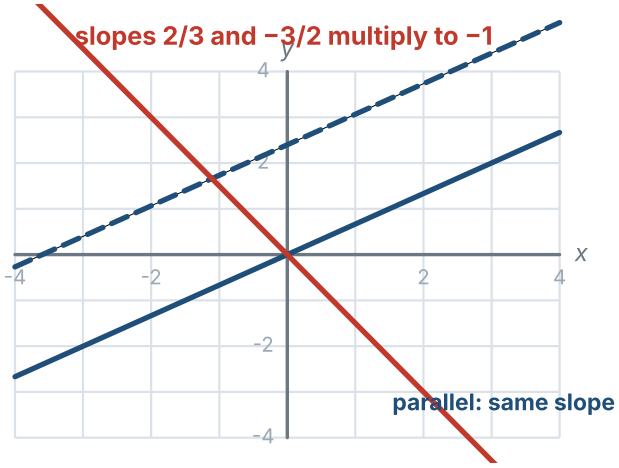
Solution.
The run is 0, so the slope is **undefined** — the line is vertical.



4 Parallel and perpendicular

parallel → equal slopes | perpendicular → slopes multiply to −1 (negative reciprocals)

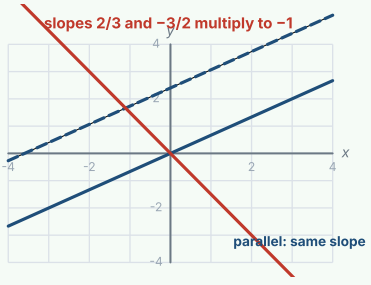
To get a perpendicular slope, flip the fraction and change the sign. The perpendicular to a slope of 2/3 is −3/2.



The two blue lines never meet — same slope 2/3. The red line crosses them at a right angle, and its slope −3/2 is the fraction flipped and negated.

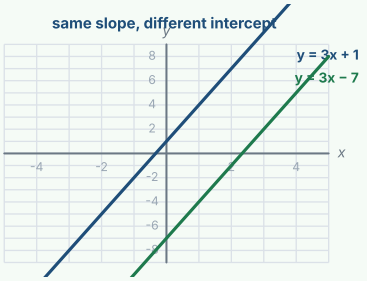
Example 1. **A line has slope 2/3. What is the slope of a line perpendicular to it?**

Solution.
Flip and negate.
−3/2.



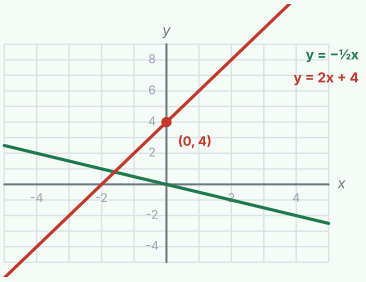
Example 2. **Is y = 3x + 1 parallel to y = 3x − 7?**

Solution.
Both have slope 3.
Yes — parallel, and never meeting.



Example 3. **A line through (0, 4) is perpendicular to y = −½x. Write its equation.**

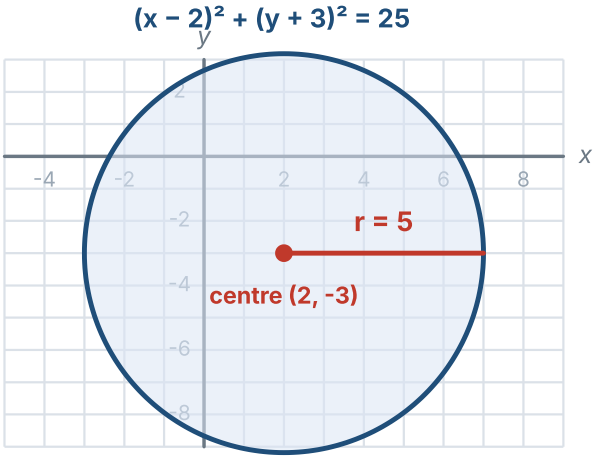
Solution.
The perpendicular slope is 2.
y = 2x + 4.



5 The equation of a circle in the plane

$$(x - h)^2 + (y - k)^2 = r^2$$

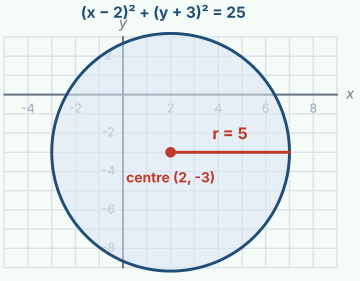
This is coordinate geometry's one memorised formula, and it is not on the reference sheet. The centre's coordinates appear with their signs flipped.



Centre (2, −3) with radius 5. In the equation the 2 appears as − 2 and the −3 appears as + 3, and the right-hand side is $5^2 = 25$.

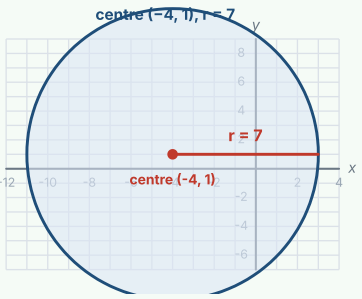
Example 1. **A circle has centre (2, −3) and radius 5. Write its equation.**

Solution.
$(x - 2)^2 + (y + 3)^2 = 25$.



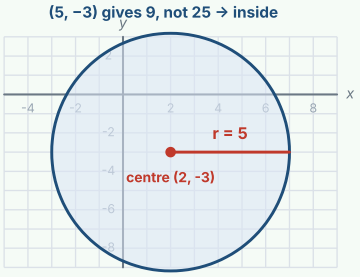
Example 2. **Find the centre and radius of $(x + 4)^2 + (y - 1)^2 = 49$.**

Solution.
Centre **(−4, 1)**, radius **7**.



Example 3. **Is (5, −3) on the circle with centre (2, −3) and radius 5?**

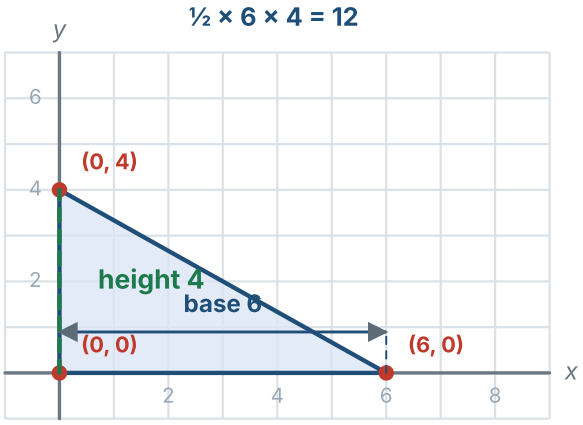
Solution.
$(5 - 2)^2 + 0 = 9$, not 25.
No — it is inside.



6 Area on a grid

for a right-angled or axis-aligned figure, read the legs straight off the grid

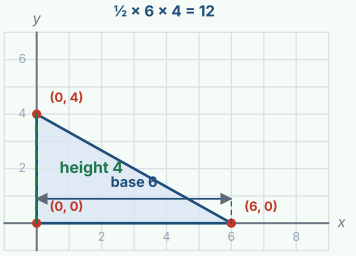
When a figure sits on the coordinate axes, you rarely need a formula beyond the basics: count the horizontal and vertical lengths and apply the ordinary area rule.



Both legs lie along the axes, so the base is 6 and the height is 4 — no distance formula needed.

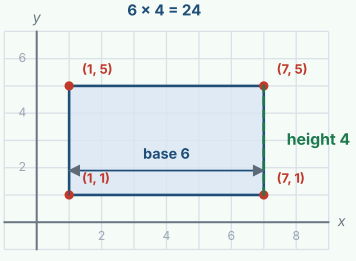
Example 1. **Find the area of a triangle with vertices (0, 0), (6, 0) and (0, 4).**

Solution.
Legs 6 and 4 along the axes.
½(6)(4) = **12**.



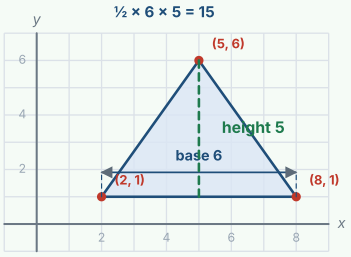
Example 2. **Find the area of a rectangle with vertices (1, 1), (7, 1), (7, 5) and (1, 5).**

Solution.
Width 6, height 4.
24.



Example 3. **Find the area of a triangle with vertices (2, 1), (8, 1) and (5, 6).**

Solution.
Base 6 along y = 1, height 5.
½(6)(5) = **15**.



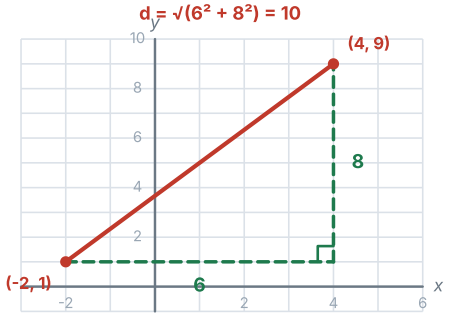
Part 1 — With Desmos (10 questions · draw the figure; the picture is faster than the algebra)

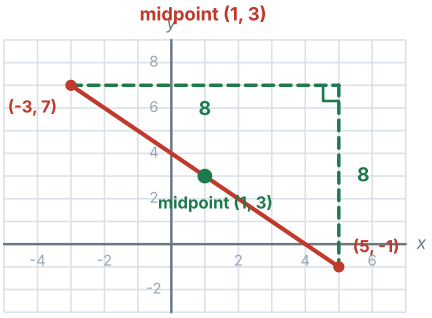
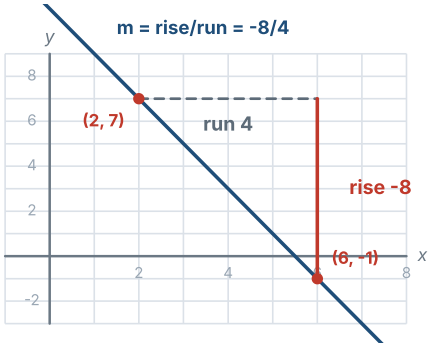
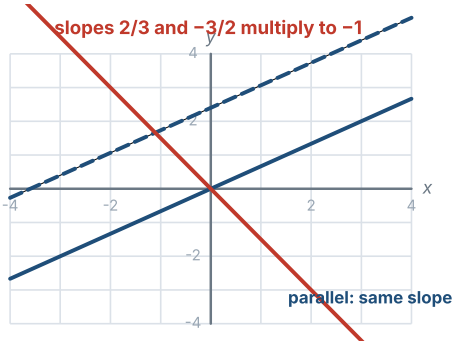
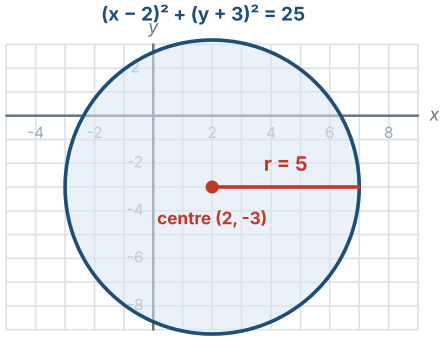
The SAT gives you Desmos on every maths question. Here it is **a plotter**. Enter the points and the line; seeing the picture settles slope signs and perpendicularity faster than any rule.

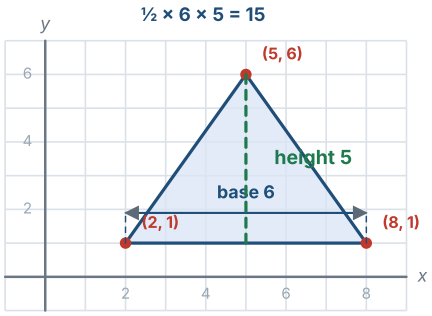
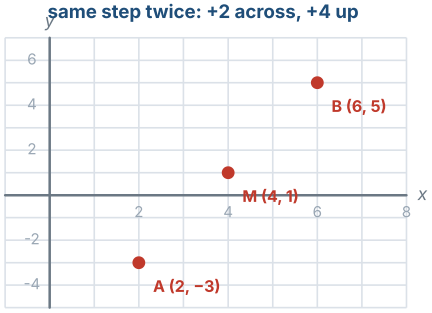
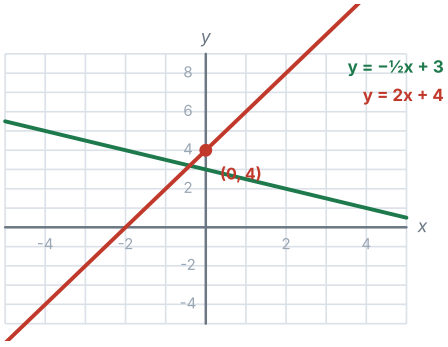
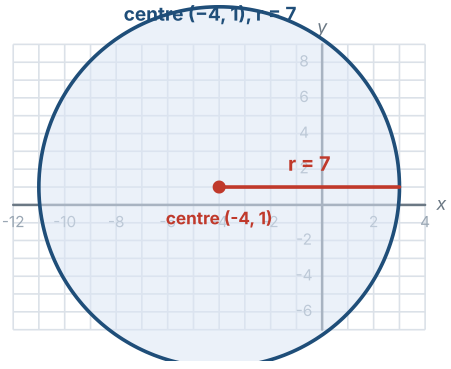
#	Question	Type into Desmos	Then	Answer
D1	What is the distance between (1, 2) and (5, 5)?	`sqrt((5-1)^2+(5-2)^2)`	Pythagoras on the two differences.	5
D2	What is the distance between (−2, 1) and (4, 9)?	`sqrt(6^2+8^2)`	A 6-8-10 triangle.	10
D3	What is the midpoint of (1, 2) and (5, 5)?	`((1+5)/2,(2+5)/2)`	Average each coordinate.	(3, 3.5)
D4	What is the slope through (2, 7) and (6, −1)?	`(-1-7)/(6-2)`	Keep the order consistent.	−2
D5	A line has slope 2/3. What is the perpendicular slope?	`-1/(2/3)`	Flip and negate.	−1.5
D6	The midpoint of AB is (4, 1) and A is (2, −3). What is the x-coordinate of B?	`2*4-2`	Twice the midpoint minus the known endpoint.	6
D7	What is the radius of $(x + 4)^2 + (y − 1)^2 = 49$?	`sqrt(49)`	The right-hand side is r squared.	7
D8	What is the area of a triangle with vertices (0, 0), (6, 0) and (0, 4)?	`0.5*6*4`	The legs lie along the axes.	12
D9	What is the area of a triangle with vertices (2, 1), (8, 1) and (5, 6)?	`0.5*6*5`	Base along y = 1, height from y = 1 to y = 6.	15
D10	Is (5, −3) on the circle centred (2, −3) with radius 5?	`(5-2)^2+(-3+3)^2`	Compare the result with r squared.	no, 9 is not 25

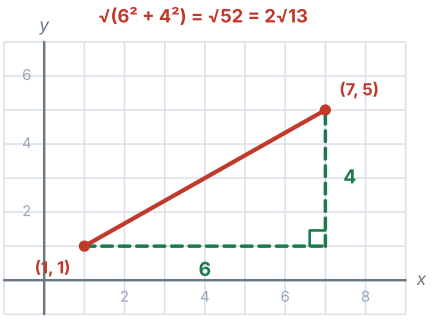
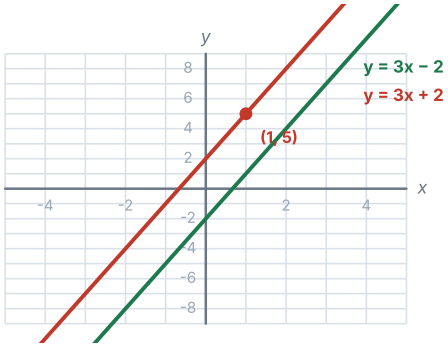
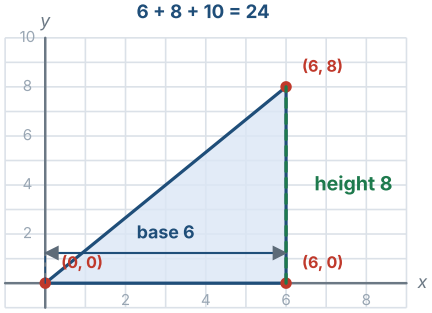
Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Sketch the points on a quick set of axes first. Nearly every question in this topic is Pythagoras, an average, or a slope — and the sketch tells you which.

#	Level	Question	Answer	Solution
P1	Hard	What is the distance between the points (−2, 1) and (4, 9) in the xy-plane? 	10	The horizontal difference is 6 and the vertical difference is 8. √(36 + 64) = √100 = **10**.

#	Level	Question	Answer	Solution
P2	Hard	What is the midpoint of the segment joining (−3, 7) and (5, −1)? 	(1, 3)	Average each coordinate: ((−3 + 5)/2, (7 − 1)/2). **(1, 3)**. A common wrong answer subtracts the coordinates instead of averaging them.
P3	Hard	What is the slope of the line passing through (2, 7) and (6, −1)? 	−2	(−1 − 7)/(6 − 2) = −8/4 = **−2**. A common wrong answer inverts the fraction, giving run over rise.
P4	Hard	A line has a slope of 2/3. What is the slope of any line perpendicular to it? 	−3/2	Perpendicular slopes are negative reciprocals: flip the fraction and change the sign. **−3/2**. A common wrong answer flips without negating; another negates without flipping.
P5	Hard	A circle has centre (2, −3) and radius 5. Write the equation of that circle. 	(x − 2)² + (y + 3)² = 25	Signs flip inside the brackets and the right side is r². **(x − 2)² + (y + 3)² = 25.** A common wrong answer uses r instead of r².

#	Level	Question	Answer	Solution
P6	Hard	A triangle has vertices (2, 1), (8, 1) and (5, 6). What is its area, in square units? 	15	The base lies along y = 1 from x = 2 to x = 8, so it is 6 long, and the height is 6 − 1 = 5. ½(6)(5) = **15**.
P7	Very hard	The midpoint of segment AB is (4, 1). If A has coordinates (2, −3), what are the coordinates of B? 	(6, 5)	The midpoint is the average, so B is the same step again: (4 + 2, 1 + 4). **(6, 5)**. A common wrong answer is the midpoint of A and the given midpoint.
P8	Very hard	Line k passes through (0, 4) and is perpendicular to the line y = −½x + 3. What is the equation of line k? 	y = 2x + 4	The perpendicular slope to −½ is **2**, and the y-intercept is given as 4. **y = 2x + 4.** A common wrong answer flips the fraction without changing the sign.
P9	Very hard	In the xy-plane, a circle has equation $(x + 4)^2 + (y − 1)^2 = 49$. What is the sum of the coordinates of its centre and its radius? 	4	Centre (−4, 1) and radius 7. −4 + 1 + 7 = **4**.

#	Level	Question	Answer	Solution
P10	Very hard	A square has two opposite vertices at (1, 1) and (7, 5). What is the length of its diagonal? 	2√13	The diagonal joins the two given points: √(36 + 16) = √52. √52 = 2√13 ≈ 7.21, so **2√13**. A common wrong answer is the same value unsimplified — the SAT expects the simplified radical.
P11	Very hard	Line m has equation y = 3x − 2. Which line is parallel to m and passes through (1, 5)? 	y = 3x + 2	Parallel means the same slope, 3, and 5 = 3(1) + b gives b = 2. **y = 3x + 2.** A common wrong answer is the original line; another does not pass through (1, 5).
P12	Very hard	Points A(0, 0), B(6, 0) and C(6, 8) form a triangle. What is its perimeter? 	24	AB = 6 and BC = 8, and the right angle at B makes AC = 10 by the 6-8-10 triple. 6 + 8 + 10 = **24**. A common wrong answer omits the hypotenuse; another is twice the area.

The one habit to build. Distance is Pythagoras; midpoint is an average. Confusing the two is the commonest error here, and a thirty-second sketch prevents it.

All 18 worked examples and all 22 answers checked numerically (see verify.py); every figure is drawn to the numbers in the question.